Trade Shocks and Political Change in Local Labor Markets

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1 Summary

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Autor, Dorn, and Hanson (2013, hereafter ADH), "The China Syndrome: Local Labor Market Effects of Import Competition in the United States," examine how rising Chinese import competition affected U.S. local labor markets between 1990 and 2007 (Autor et al. 2013). Their primary research question is whether greater exposure to Chinese imports contributed to declines in manufacturing employment, wages, and labor force participation across U.S. regions. Using commuting zones (CZs) as local labor market units, ADH use a shift-share research design that relates changes in local outcomes to differences in baseline industry composition and subsequent industry-level import growth.

Our project extends this framework to political outcomes. We ask whether CZs more exposed to the China shock experienced different changes in Republican presidential vote margins, and whether these effects appear in the short run, medium run, or long run. To answer this question, we combine the ADH replication package with county-level presidential election returns and historical county-to-CZ crosswalks. Our extension finds suggestive evidence that trade-exposed places experienced later rightward shifts in presidential voting. However, the interpretation is cautious: shift-share designs rely on nontrivial assumptions about initial industry shares and import-growth shocks, and CZ-level political change may reflect belief updating, turnout and registration, elite political supply, or population composition rather than a single mechanism.

2 Introduction

Our project's research question is does exposure to the China trade shock affect political outcomes in the United States? If so, how do these effects evolve over time? The project examines whether commuting zones that were more exposed to rising Chinese import competition experienced different changes in Republican presidential vote margins than less-exposed commuting zones. The question is not only whether trade exposure mattered politically, but also when its effects appeared. For example, we distinguish short-run effects around the Great Recession, medium-run effects around the 2016 election, and longer-run effects that may reflect later shocks such as COVID-19 and inflation.

Understanding this question is important because globalization may reshape political behavior along with the economy. If local labor-market losses from import competition contributed to changes in voting patterns, then trade policy may have broader consequences for democratic representation, polarization, and public support for protectionism. This may help explain why places hit hardest by manufacturing decline may respond differently to national political messages about trade, immigration, redistribution, and economic nationalism.

The question is also important because it connects local economic shocks to national political outcomes. A trade shock that lowers manufacturing employment may eventually influence turnout, party support, media consumption, and attitudes toward government intervention in affected regions. If these effects are persistent, then the consequences of globalization may extend beyond the original employment losses. Thus, studying these dynamics can help policymakers understand the extent of the effect of trade policy and adjustment assistance on political life.

A key empirical challenge is that identification relies on both the measurement of exposure and the validity of the shift-share design. Principally, the ADH exposure measure may suffer from aggregation and temporal bias. Import exposure is aggregated across broad industries, which may conceal substantial heterogeneity at more disaggregated levels. If variation in exposure is driven by a small subset of industries, such as textiles, furniture, and electronics, estimated effects may reflect industry-specific shocks rather than a general China-shock effect.

There are concerns regarding the validity of the Bartik instrument and initial industry shares. Identification depends not only on Chinese export growth but also on baseline industry composition in 1990. These industry shares are not randomly assigned across commuting zones and may be correlated with unobserved determinants of political outcomes, including automation exposure, unionization, education, demographic structure, or pre-existing political trends. Thus, both the IV strategy and reduced-form exposure design require careful discussion of what variation identifies the estimates.

Additionally, there is potential compositional bias due to migration across commuting zones. If trade-exposed regions experience selective in- or out-migration, observed political changes may reflect shifts in population composition instead of changes in individual political preferences. Measurement issues can also arise from mapping county-level election data into historical CZ boundaries and from sensitivity to alternative spatial definitions of labor markets. Together, these concerns motivate our robustness checks, measurement diagnostics, and cautious interpretation.

3 Data

Our analysis combines four primary datasets to construct commuting-zone-level political and trade exposure measures. We use the ADH (2013) replication package for baseline commuting-zone trade exposure measures and pre-determined labor market controls, including industry composition and demographic structure (Autor et al. 2013).

To align modern political data with historical labor markets, we use Ferrara, Testa, and Zhou (2024), *Crosswalks for U.S. Counties and Congressional Districts, 1790–2020*, which maps contemporary counties to historical boundaries and provides county centroid coordinates (Ferrara et al. 2024). Our preferred crosswalk uses M5 weights where available and M2 weights as fallback for source counties where M5 weights are undefined or zero.

We aggregate counties into 1990 commuting zones using the U.S. Department of Agriculture Economic Research Service Commuting Zones and Labor Market Areas dataset, consistent with the Tolbert and Sizer (1996) classification (Tolbert and Sizer 1996; U.S. Department of Agriculture Economic Research Service 2026).

Political outcomes are drawn from Amlani and Algara (2021), *Partisanship & Nationalization in American Elections, 1872–2020*, which provides county-level presidential election returns used to construct Republican vote margins (Amlani and Algara 2021).

We follow the sample design of Autor, Dorn, and Hanson (2013). The unit of analysis is the 1990 commuting zone (CZ) of Tolbert and Sizer (1996), covering 722 CZs over 1990–2000 and 2000–2007 stacked in first differences (1,444 observations). Table Figure 1 reports population-weighted means from ADH's replication package. Two findings stand out. Chinese imports per worker (10-year equivalent) more than doubled, from \$1.14k in 1990–2000 to \$2.63k in 2000–2007, tracking China's WTO accession in 2001. Manufacturing employment as a share of population declined from 12.69% in 1990 to 8.51% in 2007, while SSDI receipt increased over the same period.

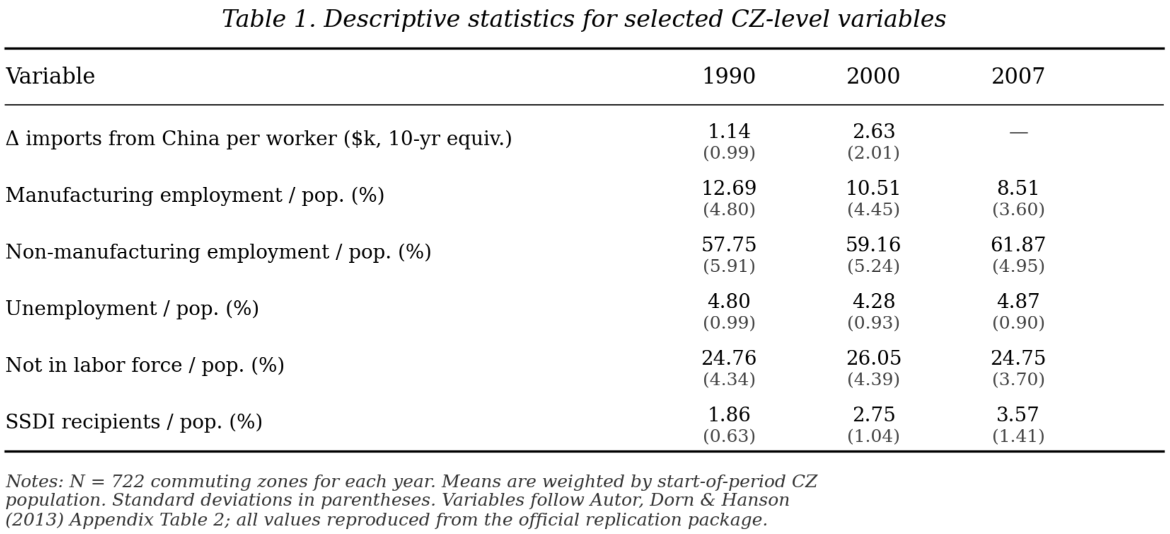

Table 1. Descriptive statistics for selected CZ-level variables

Variable	1990	2000	2007
Δ imports from China per worker (\$k, 10-yr equiv.)	1.14 (0.99)	2.63 (2.01)	—
Manufacturing employment / pop. (%)	12.69 (4.80)	10.51 (4.45)	8.51 (3.60)
Non-manufacturing employment / pop. (%)	57.75 (5.91)	59.16 (5.24)	61.87 (4.95)
Unemployment / pop. (%)	4.80 (0.99)	4.28 (0.93)	4.87 (0.90)
Not in labor force / pop. (%)	24.76 (4.34)	26.05 (4.39)	24.75 (3.70)
SSDI recipients / pop. (%)	1.86 (0.63)	2.75 (1.04)	3.57 (1.41)

Notes: N = 722 commuting zones for each year. Means are weighted by start-of-period CZ population. Standard deviations in parentheses. Variables follow Autor, Dorn & Hanson (2013) Appendix Table 2; all values reproduced from the official replication package.

Figure 1: Descriptive statistics for selected CZ-level variables, reproduced from ADH 2013 Appendix Table 2.

4 Analysis and Findings

4.1 Identification Strategy and its Plausibility

OLS estimation of the labor-market effect of Chinese import exposure faces an endogeneity problem. Observed U.S. imports reflect both Chinese supply-side and U.S. demand shocks, biasing OLS toward zero. ADH (2013) report an OLS estimate of −0.171 versus an IV estimate of −0.596 (Autor et al. 2013).

They address this with a Bartik-style instrument, replacing ΔIPW_{uit} with ΔIPW_{oit}, built from same-period Chinese exports to eight other high-income countries (Australia, Denmark, Finland, Germany, Japan, New Zealand, Spain, and Switzerland). The identifying assumption is that within-industry import growth shared by these countries reflects China's supply-side improvements, not correlated demand:

$$\Delta IPW_{uit} = \sum_j \left(\frac{L_{ijt}}{L_{ujt}}\right) \cdot \left(\frac{\Delta M_{ucjt}}{L_{it}}\right),$$

$$\Delta IPW_{oit} = \sum_j \left(\frac{L_{ij,t-1}}{L_{uj,t-1}}\right) \cdot \left(\frac{\Delta M_{ocjt}}{L_{i,t-1}}\right).$$

The second-stage equation is estimated as stacked first differences:

$$\Delta L_{itm} = \gamma_t + \beta_1 \Delta IPW_{uit} + X'_{it}\beta_2 + e_{it},$$

where X_{it} contains start-of-period manufacturing share, demographic controls, the routine and offshorability indices of Autor and Dorn (2013), and Census division dummies (Autor and Dorn 2013).

Three pieces of evidence support the strategy. First, the first stage is strong: the instrument predicts the endogenous regressor with coefficient 0.631 ($t = 6.90$), well above the weak-IV threshold. Second, a falsification test shows that 1970s–1980s manufacturing changes do not predict future Chinese exposure, ruling out a pre-existing decline. Third, dropping suspect industries leaves the coefficient near -0.596.

The main residual concern is correlated demand shocks across high-income economies. A gravity-based check yields similar estimates; Chinese manufacturing TFP grew about 8% per year over 1998–2007, against 3.9% in the U.S., supporting the supply-driven interpretation.

4.2 Reproduction of Published Results

We verify that the official replication package functions correctly. Our target is Table 3 of ADH (2013), the 2SLS regression of CZ-level change in manufacturing employment share on ΔIPW_{uit}, instrumented by the eight-country measure. We focus on column (6) with full controls.

Running the official Stata code on the published workfile, we obtain coefficients matching the published values exactly to three decimal places across all six columns. In column (6), the coefficient on ΔIPW is -0.596 (SE 0.099); the first-stage coefficient is 0.631 ($t = 6.90$); $N = 1{,}444$ with $R^2 = 0.343$. Table Figure 2 reports the full comparison.

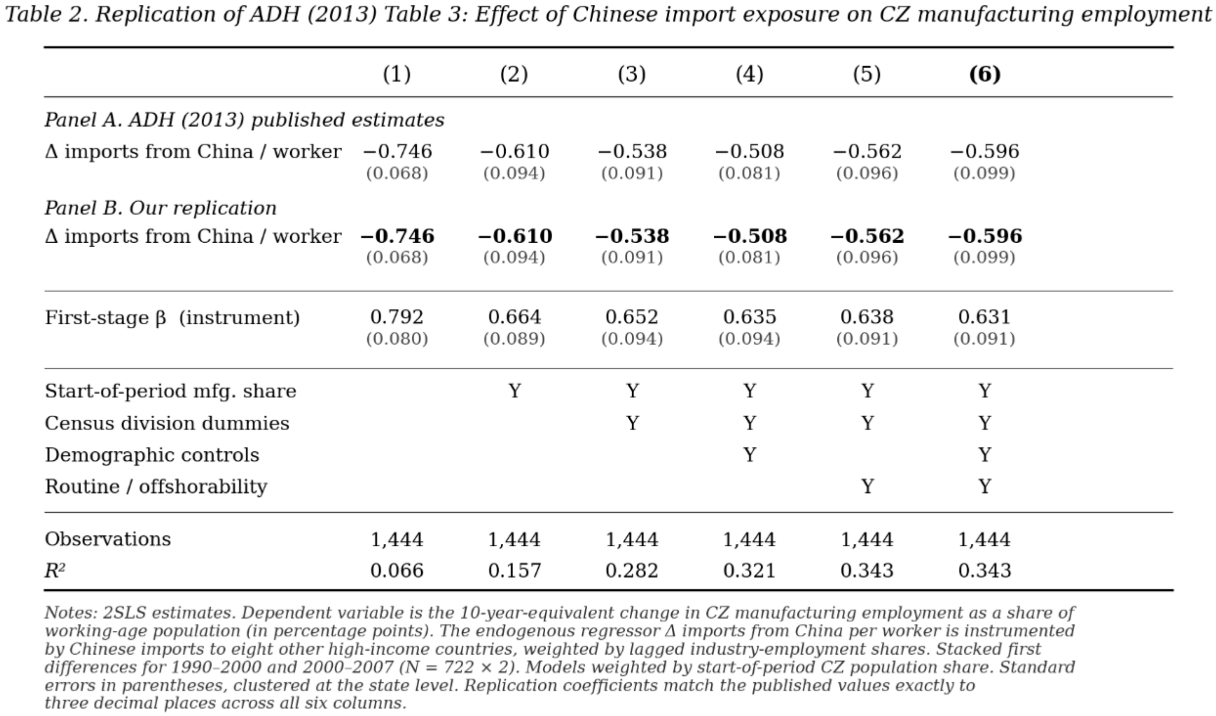

Table 2. Replication of ADH (2013) Table 3: Effect of Chinese import exposure on CZ manufacturing employment

	(1)	(2)	(3)	(4)	(5)	**(6)**
Panel A. ADH (2013) published estimates						
Δ imports from China / worker	−0.746	−0.610	−0.538	−0.508	−0.562	−0.596
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
Panel B. Our replication						
Δ imports from China / worker	**−0.746**	**−0.610**	**−0.538**	**−0.508**	**−0.562**	**−0.596**
	(0.068)	(0.094)	(0.091)	(0.081)	(0.096)	(0.099)
First-stage β (instrument)	0.792	0.664	0.652	0.635	0.638	0.631
	(0.080)	(0.089)	(0.094)	(0.094)	(0.091)	(0.091)
Start-of-period mfg. share		Y	Y	Y	Y	Y
Census division dummies			Y	Y	Y	Y
Demographic controls				Y		Y
Routine / offshorability					Y	Y
Observations	1,444	1,444	1,444	1,444	1,444	1,444
R^2	0.066	0.157	0.282	0.321	0.343	0.343

Notes: 2SLS estimates. Dependent variable is the 10-year-equivalent change in CZ manufacturing employment as a share of working-age population (in percentage points). The endogenous regressor Δ imports from China per worker is instrumented by Chinese imports to eight other high-income countries, weighted by lagged industry-employment shares. Stacked first differences for 1990–2000 and 2000–2007 (N = 722 × 2). Models weighted by start-of-period CZ population share. Standard errors in parentheses, clustered at the state level. Replication coefficients match the published values exactly to three decimal places across all six columns.

Figure 2: Replication of ADH (2013) Table 3 — effect of Chinese import exposure on CZ manufacturing employment share.

Economically, −0.596 implies that a $1,000 per-worker increase in decadal Chinese exposure lowers a CZ's manufacturing share by 0.6 percentage points. ADH attribute 44% of the 1990–2007 manufacturing decline to this channel. Because we use the original data and code, this is a computational reproduction—the exact match confirms our codebase functions correctly. The next section applies the same strategy to CZ-level presidential political outcomes.

4.3 Extension: Dynamic Political Effects of the China Shock

We extend the ADH local-labor-market design to presidential political outcomes. The outcome is the Republican presidential vote margin in CZ c and election year t. The preferred specification includes CZ fixed effects, election-year fixed effects, the ADH 1990–2007 import-exposure measure interacted with election-year indicators, and time-varying effects of baseline controls. The omitted year is 1988, so coefficients describe how the relationship between China-shock exposure and Republican margin changes relative to the pre-shock election immediately before the main exposure period.

A central measurement challenge is that electoral data are observed for counties whose boundaries change over time, while ADH exposure is measured for 1990 commuting zones. We therefore bridge county election returns to 1990 counties using the FTZ county crosswalks, aggregate counties to 1990 CZs using the Tolbert-Sizer/USDA crosswalk, and merge those outcomes to ADH exposure. Our preferred bridge uses FTZ M5 weights where available and M2 fallback weights where M5 is undefined or zero. We validate retained vote shares, document fallback counties, and compare the results to M6+fallback, pure M2, and M4+fallback. These checks are important because the estimates should not be driven by arbitrary geographic harmonization choices.

The empirical exercise should be interpreted as a reduced-form exposure design, not as definitive evidence of individual belief updating. ADH exposure is a shift-share measure: CZs differ in initial industry composition, and industries differ in later import growth. Identification requires either that initial industry shares are conditionally unrelated to counterfactual political trends, or that the industry-level import shifts are as-good-as-random with respect to local political trends (Goldsmith-Pinkham et al. 2020; Borusyak et al. 2025). These assumptions are not innocuous. In our diagnostics, the ADH instrument strongly predicts exposure, but exposure is also highly correlated with baseline manufacturing intensity and routine-task concentration. Table Table 1 summarizes the first-stage strength in our political-outcome sample.

Table 1: First-stage diagnostics for ADH exposure.

Specification	Estimate	Std. error	First-stage F	N
Minimal	0.977	0.085	133.4	722
Core controls	0.812	0.101	64.6	722
Full controls	0.794	0.101	62.0	722

Figure Figure 3 presents the main event-study estimates. The preferred inference is CZ-clustered. Conley, Driscoll-Kraay, Newey-West, state-clustered, and two-way clustered alternatives are computed as diagnostics, but several spatial and two-way variance-covariance matrices are non-positive-definite or require numerical repair in this short election-year panel. We therefore state the conclusions cautiously and base the main figure on CZ-clustered confidence intervals.

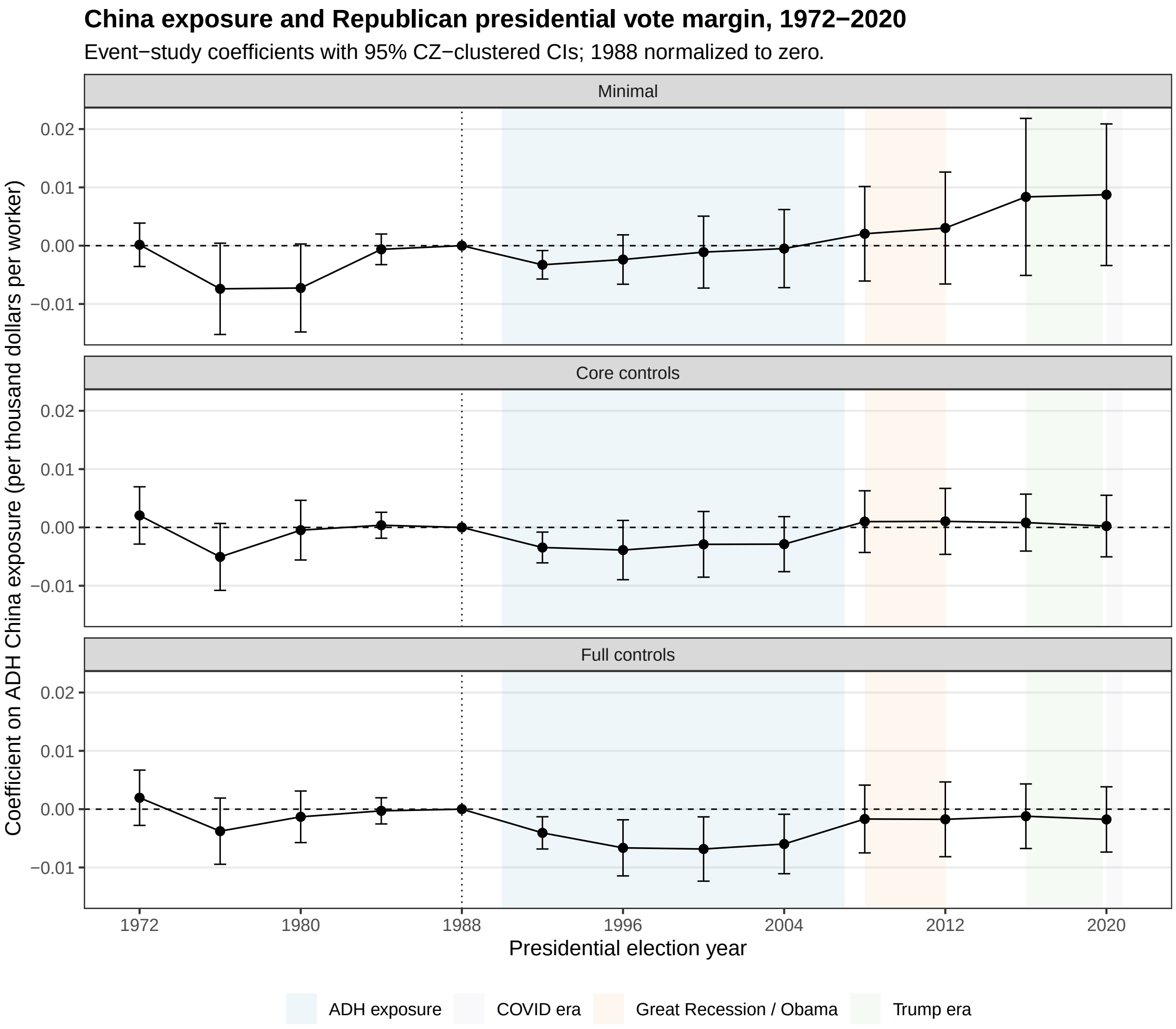

Figure 3: Event-study coefficients for ADH exposure and Republican presidential margin.

The estimates suggest little clear evidence of rising Republican margins before the shock. The post-shock pattern is more consistent with a later political response: the positive association grows in the medium and long run, especially around the 2016 and 2020 horizons. This timing is important because it suggests that the political effects of local trade exposure may not operate immediately through local material losses alone. They may also depend on later national political rhetoric, elite positioning, media narratives, or changing turnout and party coalitions.

To understand what is behind the margin effect, we decompose the outcome into Republican votes per 1990 population, Democratic votes per 1990 population, and two-party vote volume. Figure Figure 4 shows that the margin pattern should not be interpreted mechanically as a simple increase in Republican votes. The broader decomposition helps distinguish vote-choice changes from changes in participation and party-specific mobilization.

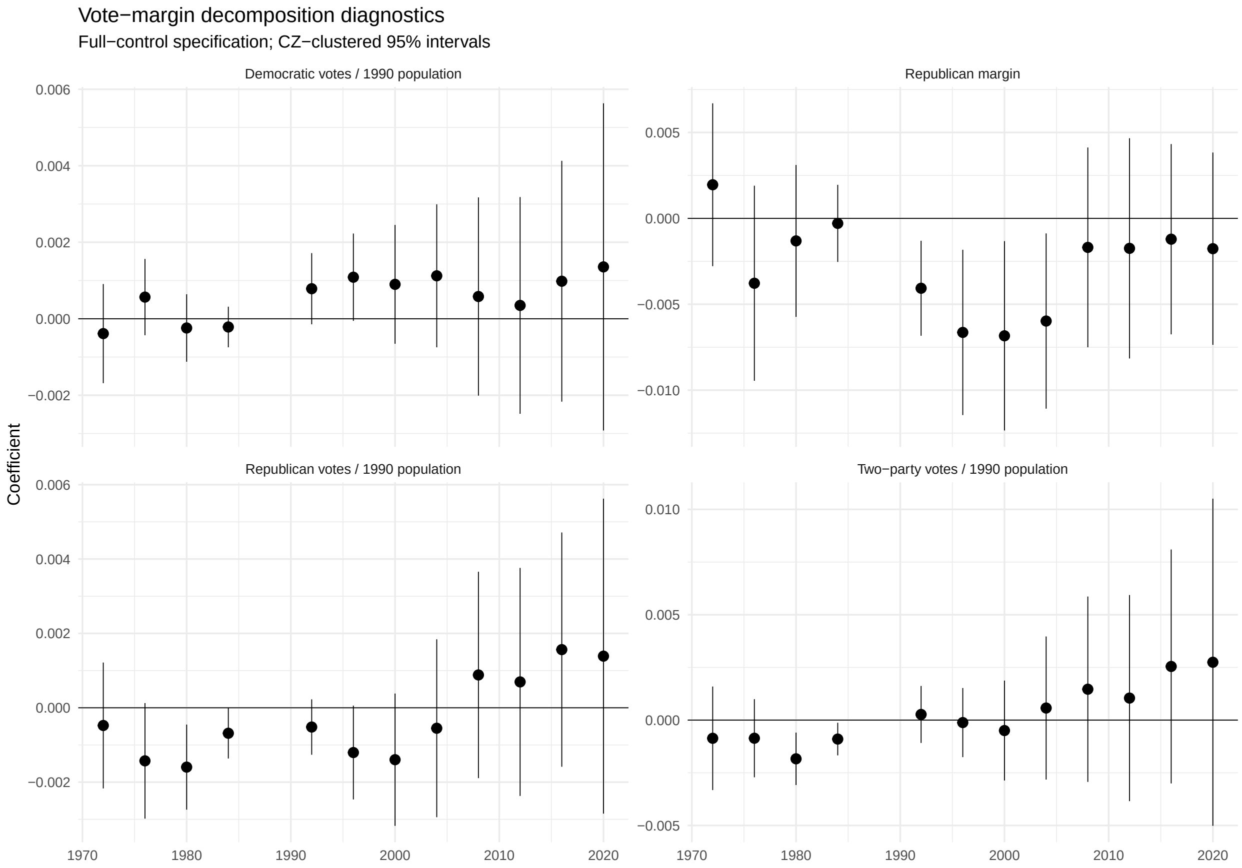

Figure 4: Political-outcome decomposition across election years.

Finally, we examine whether the single 1990–2007 exposure measure masks important timing differences. The original ADH exposure measure aggregates the China shock over a long period, so an event-study coefficient for 2016 does not mean that a new trade shock occurs in 2016. It means that CZs with greater cumulative exposure by 2007 evolved differently by 2016. To make this interpretation more transparent, we split exposure into 1990–2000 and 2000–2007 components and include both components jointly. Figure Figure 5 shows that the estimated relationship varies depending on which subperiod of import growth is emphasized. Pre-1988 coefficients in this exercise are placebo relationships with future exposure, while post-2008 coefficients describe how early- and late-shock exposure predict later political differences.

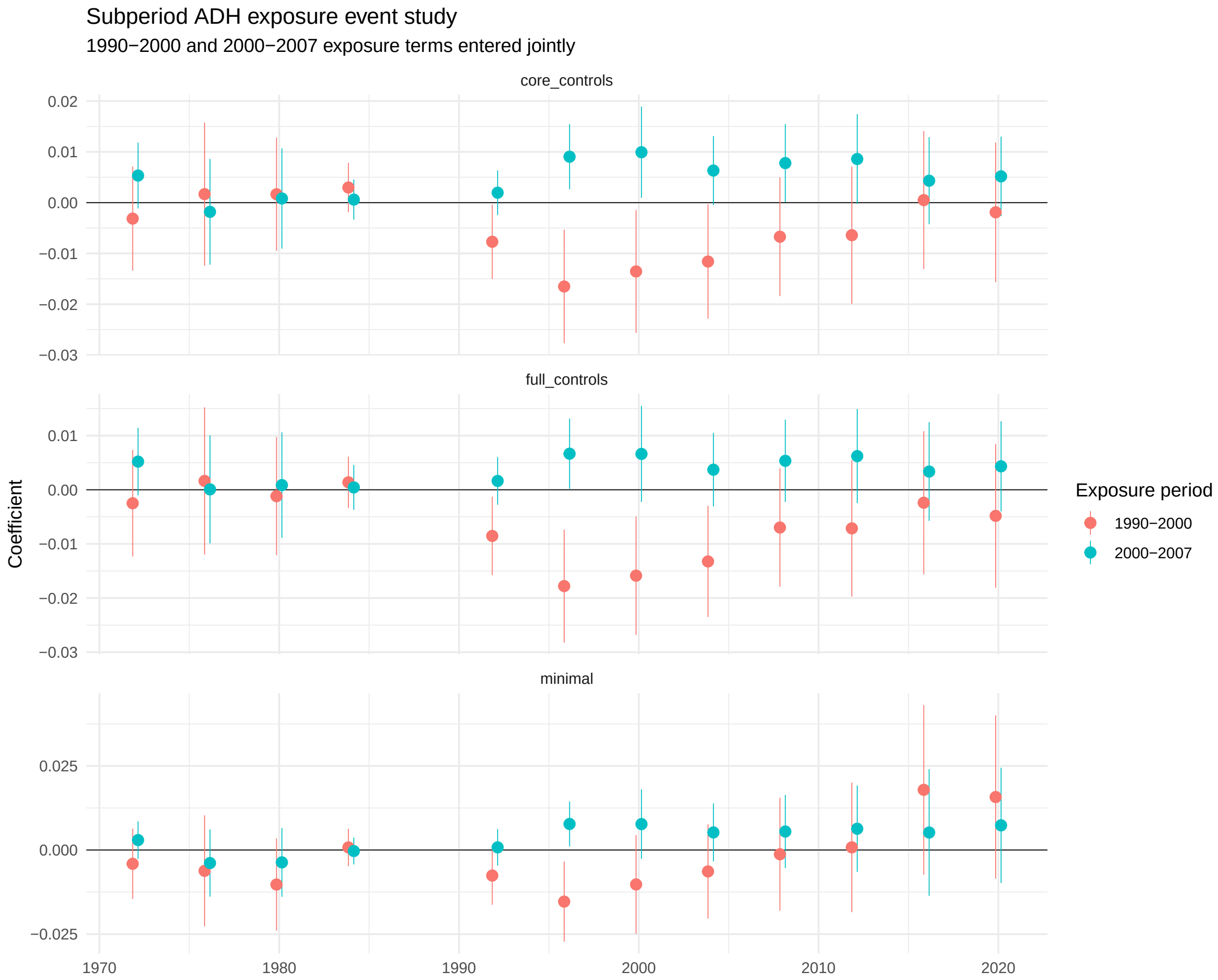

Figure 5: Subperiod ADH exposure event study, with 1990–2000 and 2000–2007 exposure terms entered jointly.

5 Conclusion

Our results suggest that the China shock may have had political consequences beyond its direct effects on manufacturing employment. CZs more exposed to Chinese import competition eventually show more Republican-leaning presidential outcomes, with the clearest differences appearing after the original labor-market shock rather than immediately. This timing makes the interpretation complex. The estimates may reflect voters responding to material local losses, but they may also reflect later elite rhetoric, media framing, turnout changes, candidate supply, or changes in local population composition.

The main lesson is that trade policy cannot be evaluated only through employment and wage effects. Large local labor-market shocks may also affect political representation, polarization, and support for future policy regimes. At the same time, our conclusions should be cautious. The design builds on a Bartik-style exposure measure, so interpretation depends on assumptions about initial industry composition and industry-level shocks. The unit of analysis is a commuting zone, not an individual voter, so we cannot directly observe individual belief updating. Our evidence is most informative about how local political outcomes evolve in places more exposed to import competition, not about a single definitive psychological mechanism.

Future work should therefore focus on mechanism. Turnout and registration data, partisan-index data, migration data, and measures of elite political supply could help distinguish material-experience channels from mobilization, composition, and rhetoric/media channels. Even with these limitations, the evidence suggests that the political effects of globalization are dynamic, geographically concentrated, and potentially mediated by institutions and narratives long after the initial economic shock.

6 References

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A Appendix A: County-to-CZ Harmonization and Measurement

This appendix documents the county-to-CZ bridge used to construct political outcomes. County election returns are bridged to 1990 counties using FTZ crosswalk weights and then aggregated to 1990 CZs. The preferred design uses M5 where valid and M2 fallback where M5 is undefined or zero.

Table 2: Pipeline validation checks.

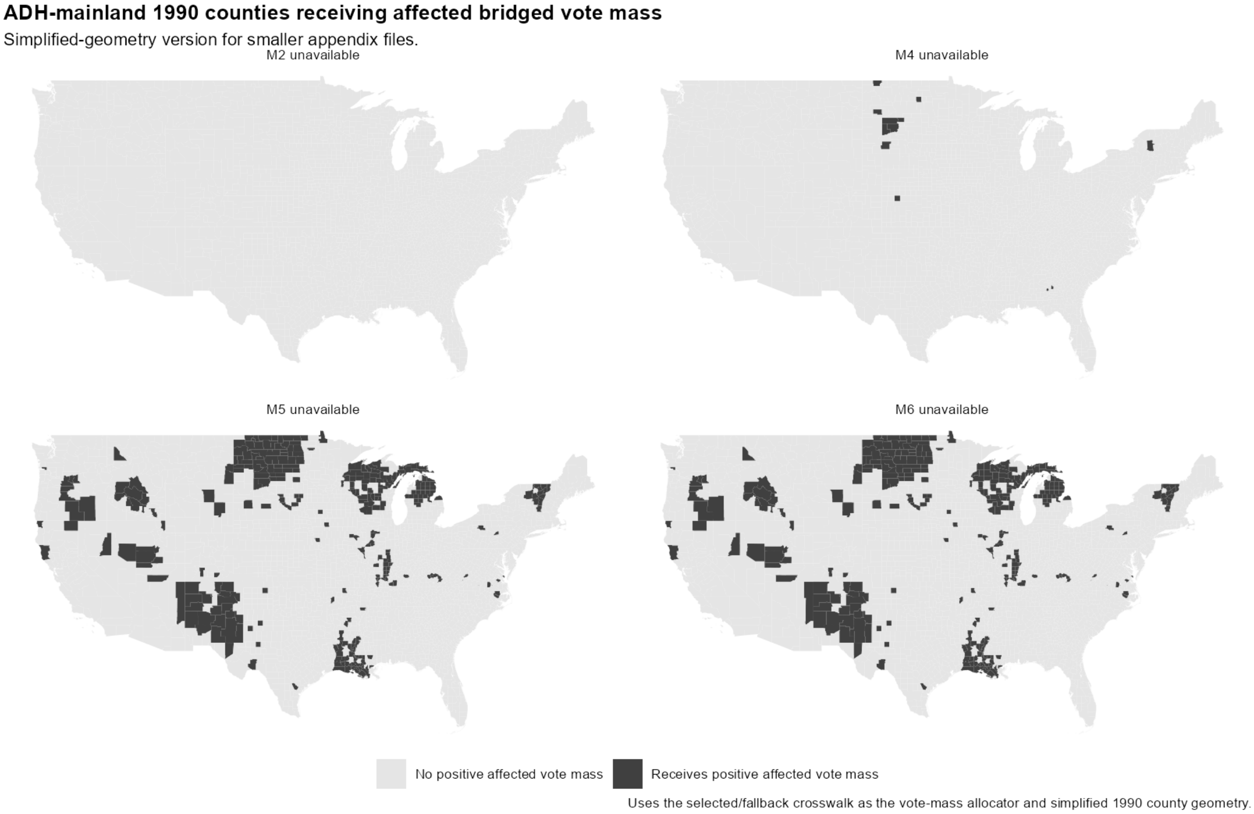
check	status	fatal	n_failed	n_reported	details
adh_one_row_per_cz	pass	TRUE	0	NA	Requires one ADH baseline/exposure row per commuting zone.
cz_lookup_unique_1990_counties	pass	TRUE	0	NA	Requires each 1990 county FIPS to appear once and map to one CZ.
aa_all_county_fips_valid	pass	TRUE	0	NA	Requires all county FIPS to be five numeric digits before filtering.
aa_vote_identity_error_within_tolerance	pass	TRUE	0	NA	Max absolute vote identity error: 0
crosswalk_positive_final_weight_sums_equal_one	pass	TRUE	0	NA	Tolerance: 1e-06
crosswalk_all_source_weight_sums_equal_one_or_reported	reported	FALSE		162	Global non-ADH or unsupported source-county rows are reported rather than treated as validation failures. Estimation is gated by positive final weights, retained vote share, and panel balance. Tolerance: 1e-06
crosswalk_undefined_selected_weights_handled	pass	TRUE	1204	NA	Policy: fallback_m2
aa_no_duplicate_county_years_after_exact_distinct	pass	TRUE	0	NA	Exact duplicate rows removed: 0
aa_two_party_votes_match_party_votes	pass	TRUE	0	NA	Requires republican + democratic votes to equal two-party votes.
aa_votes_nonnegative	pass	TRUE	0	NA	Requires nonnegative Republican, Democratic, two-party, and total votes.
bridged_vote_retention_above_threshold	pass	TRUE	0	NA	Minimum retained share in main years: 0.996122
main_panel_balanced	pass	TRUE	0	NA	Observed complete rows: 9386; expected rows: 9386; missing share: 0
event_study_residualized_design_full_rank	pass	TRUE	0	NA	Requires no residualized RHS rank deficiency in 1952-start or 1972-start specifications.
event_study_main_vcov_available_without_repair	pass	TRUE	0	NA	Main SE type: cluster_cz
event_study_conley_repair_diagnostics_recorded	pass	FALSE	31	NA	Conley VCOVs requiring repair across all samples/specs/cutoffs: 31

Table 3: Retained vote shares by year and source scope.

Year	Source counties	Fallback counties	Unmatched counties	Retained vote share
1972	3106	258	0	1.0000
1976	3109	258	0	1.0000
1980	3108	221	0	1.0000
1984	3110	221	0	1.0000
1988	3110	221	0	1.0000
1992	3110	0	0	1.0000

Table 3: Retained vote shares by year and source scope.

Year	Source counties	Fallback counties	Unmatched counties	Retained vote share
1996	3109	0	0	1.0000
2000	3109	196	0	1.0000
2004	3109	196	0	1.0000
2008	3109	196	0	1.0000
2012	3109	183	0	1.0000
2016	3108	183	0	1.0000
2020	3108	184	0	1.0000



ADH-mainland 1990 counties receiving affected bridged vote mass

Simplified-geometry version for smaller appendix files.

M2 unavailable

M4 unavailable

M5 unavailable

M6 unavailable

Uses the selected/fallback crosswalk as the vote-mass allocator and simplified 1990 county geometry.

Figure 6: 1990 counties receiving positive vote mass from source counties with unavailable weights.

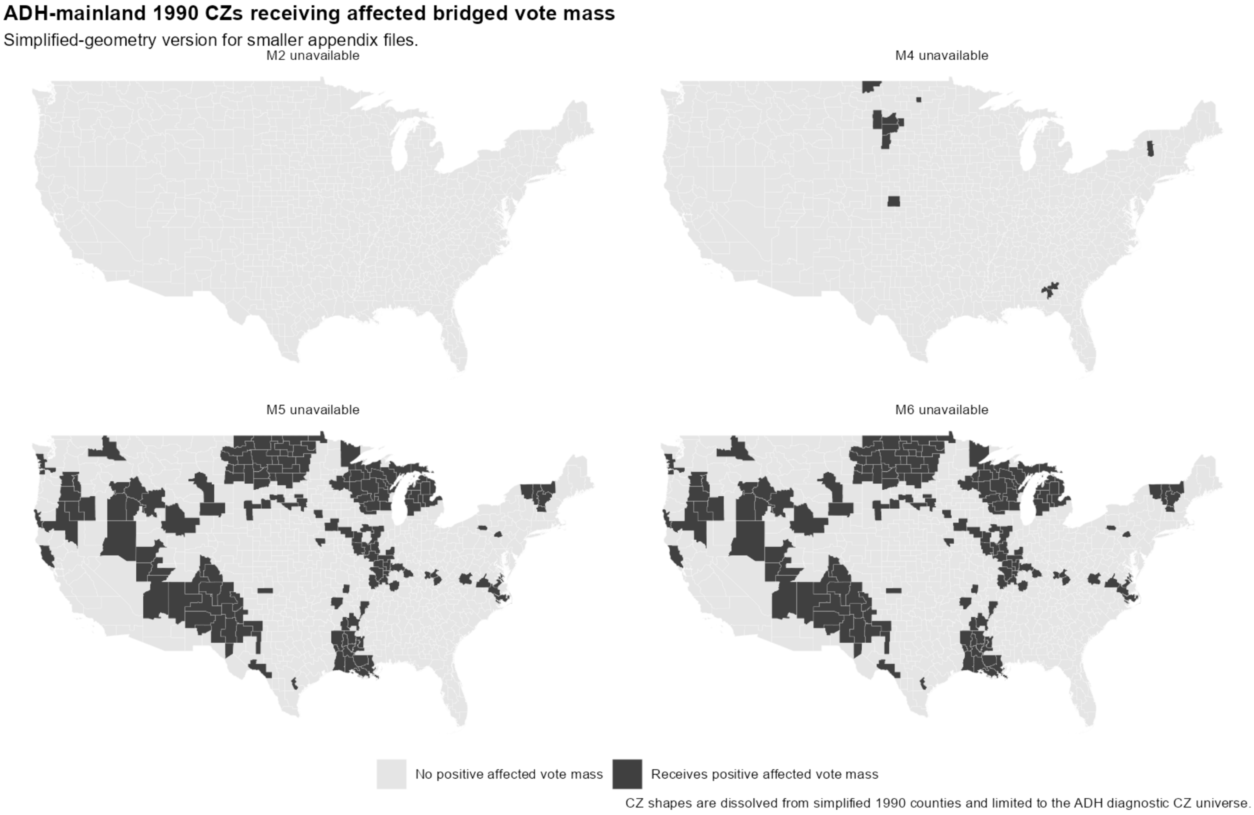


Figure 7: 1990 CZs receiving positive vote mass from source counties with unavailable weights.

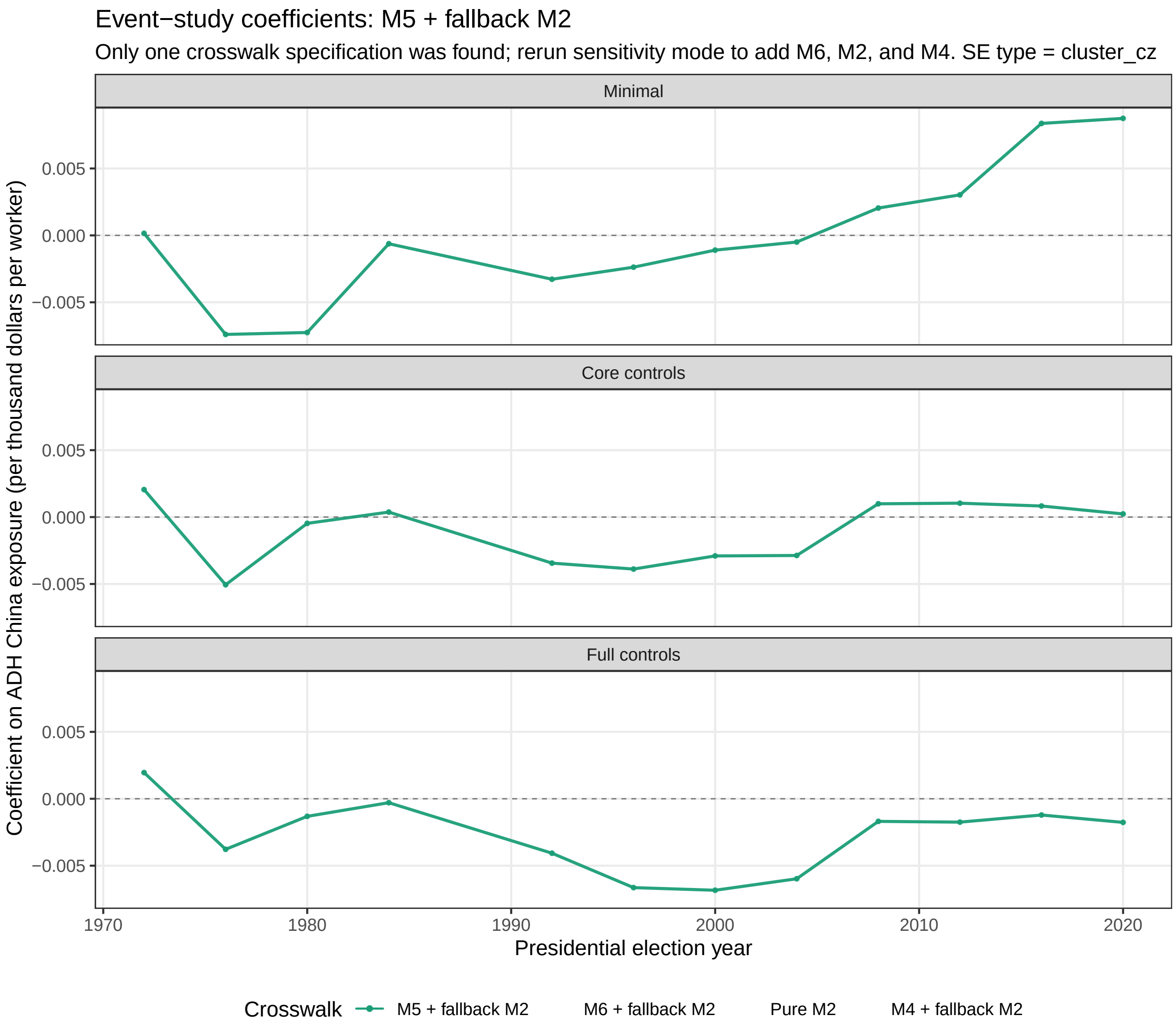


Figure 8: Crosswalk-specification comparison. If the full sensitivity pipeline has not been run, this figure will show only the preferred M5+fallback design.

B Appendix B: Main Event Study and Alternative Political Outcomes

Table 4: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Minimal	1972	0.0002	0.0019	[-0.0036, 0.0039]	0.936
Minimal	1992	-0.0033	0.0012	[-0.0057, -0.0008]	0.009
Minimal	2000	-0.0011	0.0031	[-0.0073, 0.0051]	0.726
Minimal	2008	0.0020	0.0041	[-0.0061, 0.0101]	0.621
Minimal	2016	0.0084	0.0069	[-0.0051, 0.0218]	0.224
Minimal	2020	0.0087	0.0062	[-0.0034, 0.0209]	0.159

Table 4: Main event-study coefficient table, selected years.

Spec	Year	Estimate	Std. error	95% CI	p-value
Core controls	1972	0.0021	0.0025	[-0.0029, 0.0070]	0.412
Core controls	1992	-0.0034	0.0013	[-0.0061, -0.0008]	0.011
Core controls	2000	-0.0029	0.0029	[-0.0085, 0.0027]	0.313
Core controls	2008	0.0010	0.0027	[-0.0043, 0.0063]	0.713
Core controls	2016	0.0008	0.0025	[-0.0041, 0.0057]	0.740
Core controls	2020	0.0002	0.0027	[-0.0050, 0.0055]	0.930
Full controls	1972	0.0020	0.0024	[-0.0028, 0.0067]	0.419
Full controls	1992	-0.0041	0.0014	[-0.0068, -0.0013]	0.004
Full controls	2000	-0.0068	0.0028	[-0.0123, -0.0013]	0.015
Full controls	2008	-0.0017	0.0030	[-0.0075, 0.0041]	0.569
Full controls	2016	-0.0012	0.0028	[-0.0068, 0.0043]	0.668
Full controls	2020	-0.0018	0.0029	[-0.0074, 0.0038]	0.537

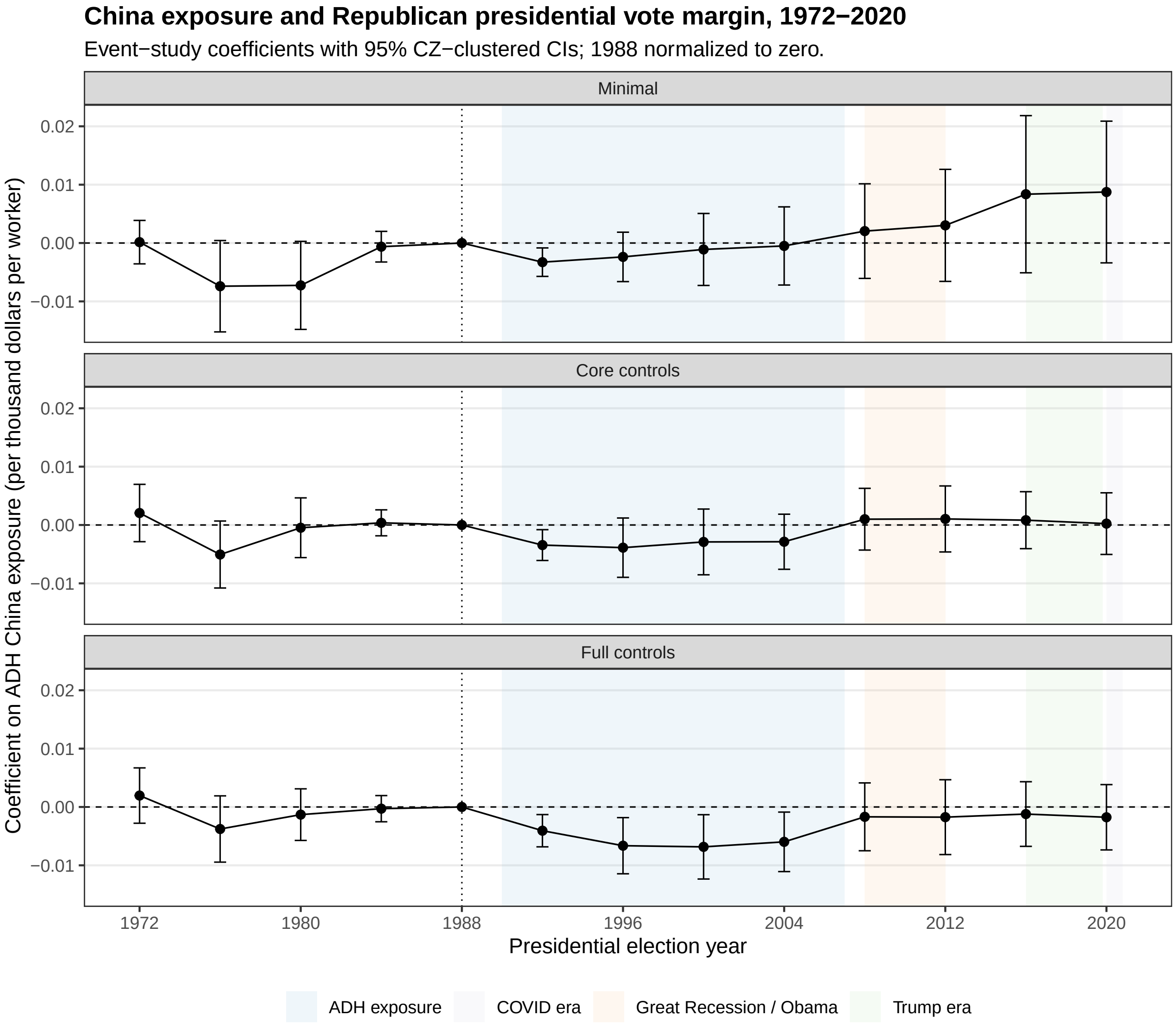
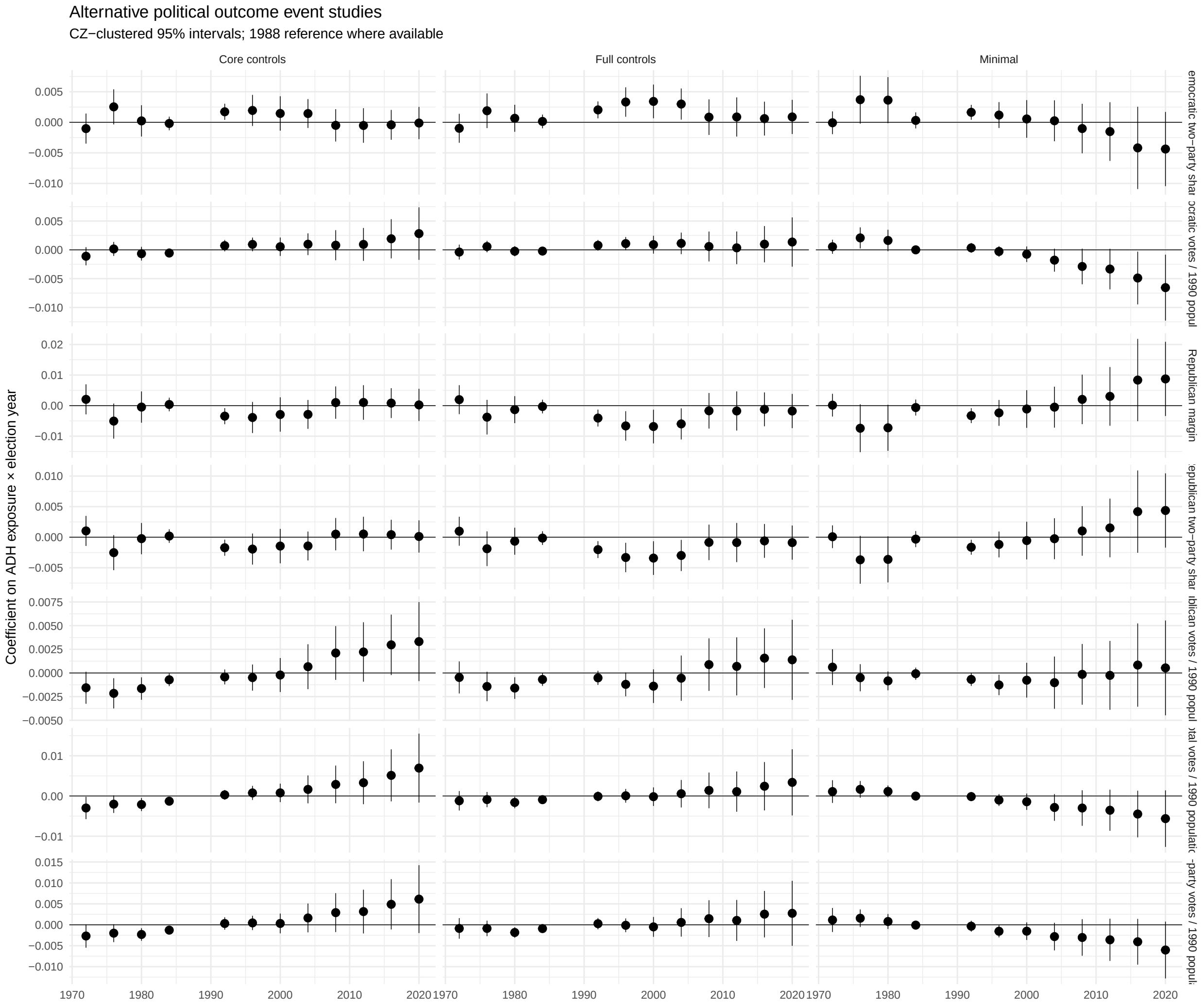


Figure 9: Main event-study estimates.



Alternative political outcome event studies

CZ−clustered 95% intervals; 1988 reference where available

Figure 10: Full alternative-outcome event-study grid.

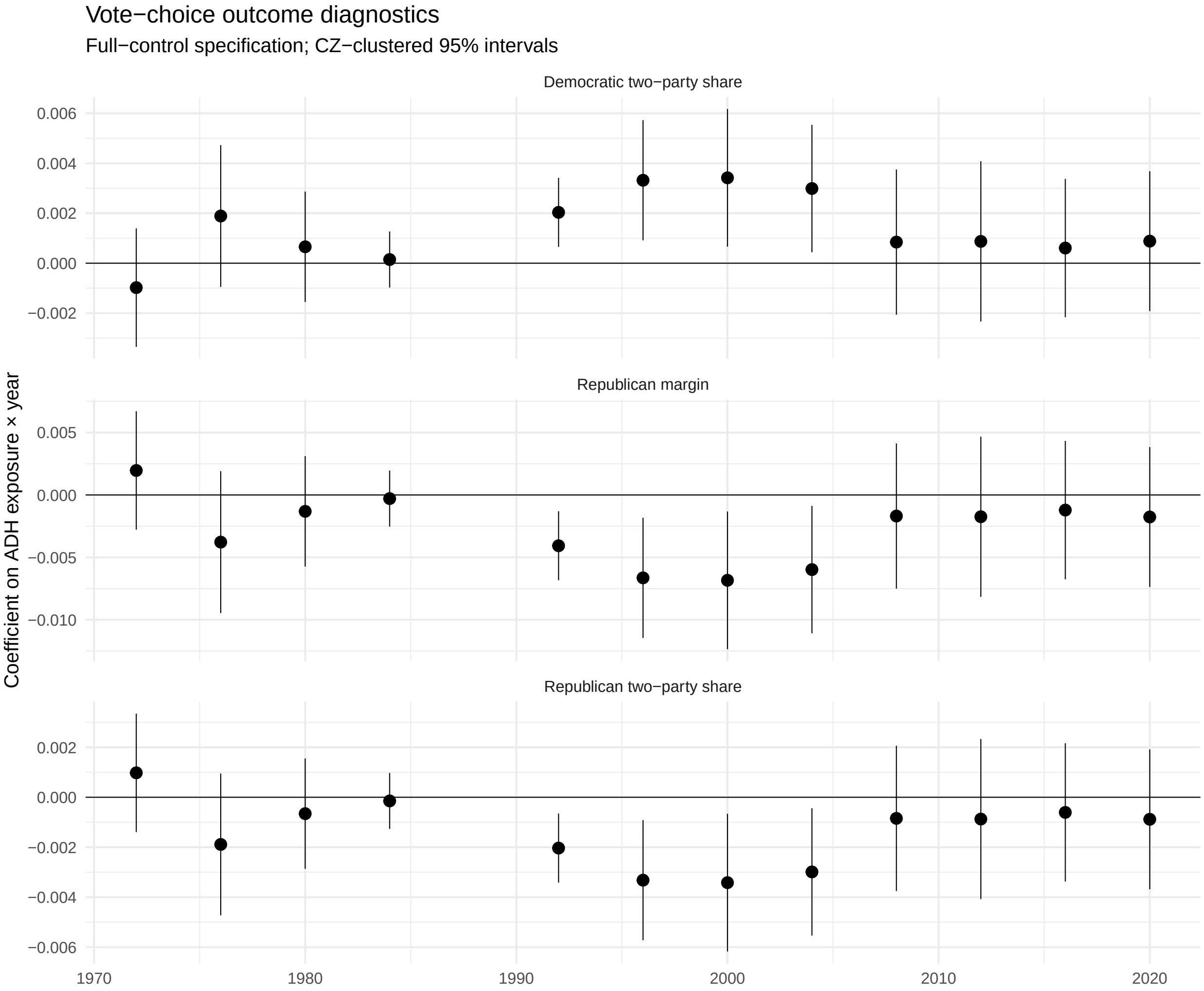


Figure 11: Vote-choice outcome family.

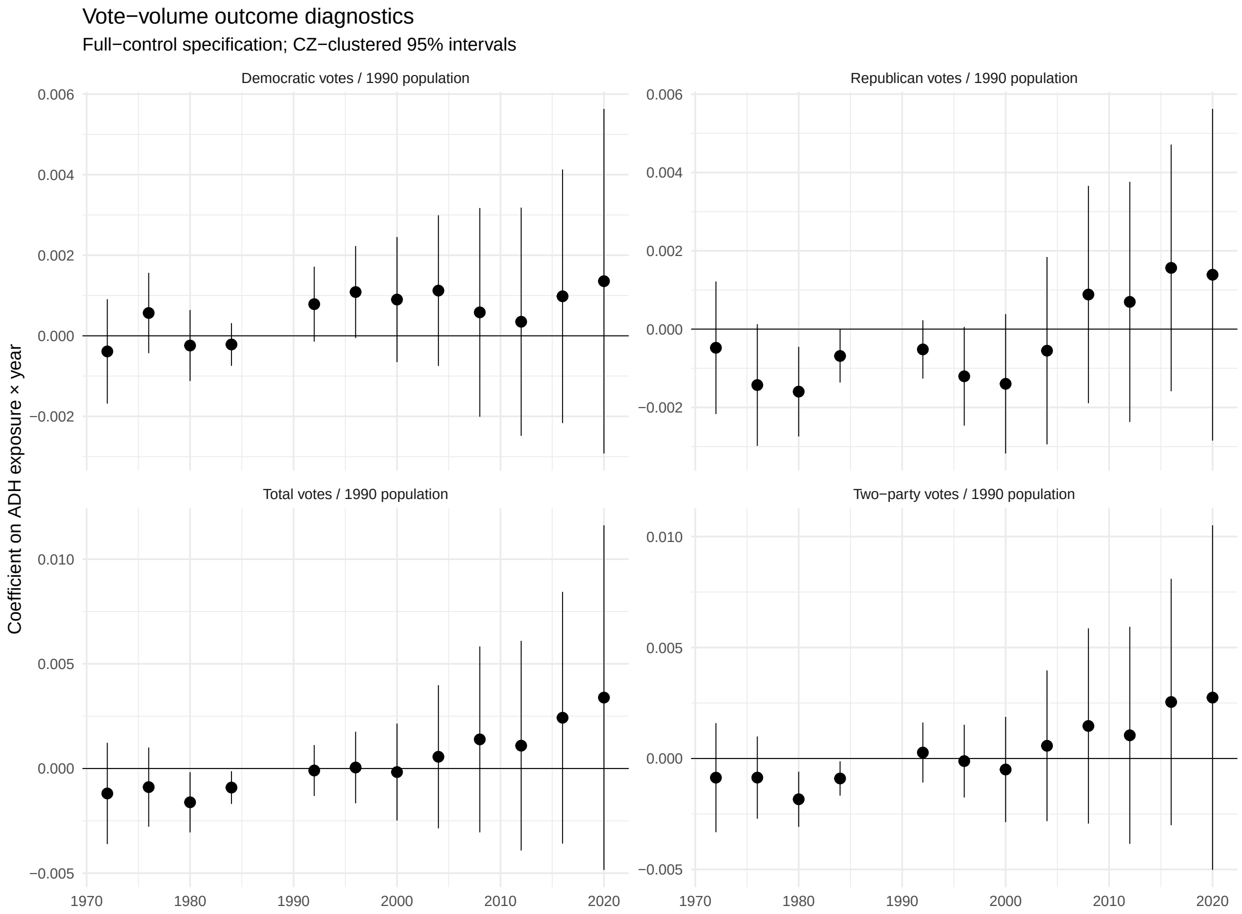


Figure 12: Vote-volume outcome family.

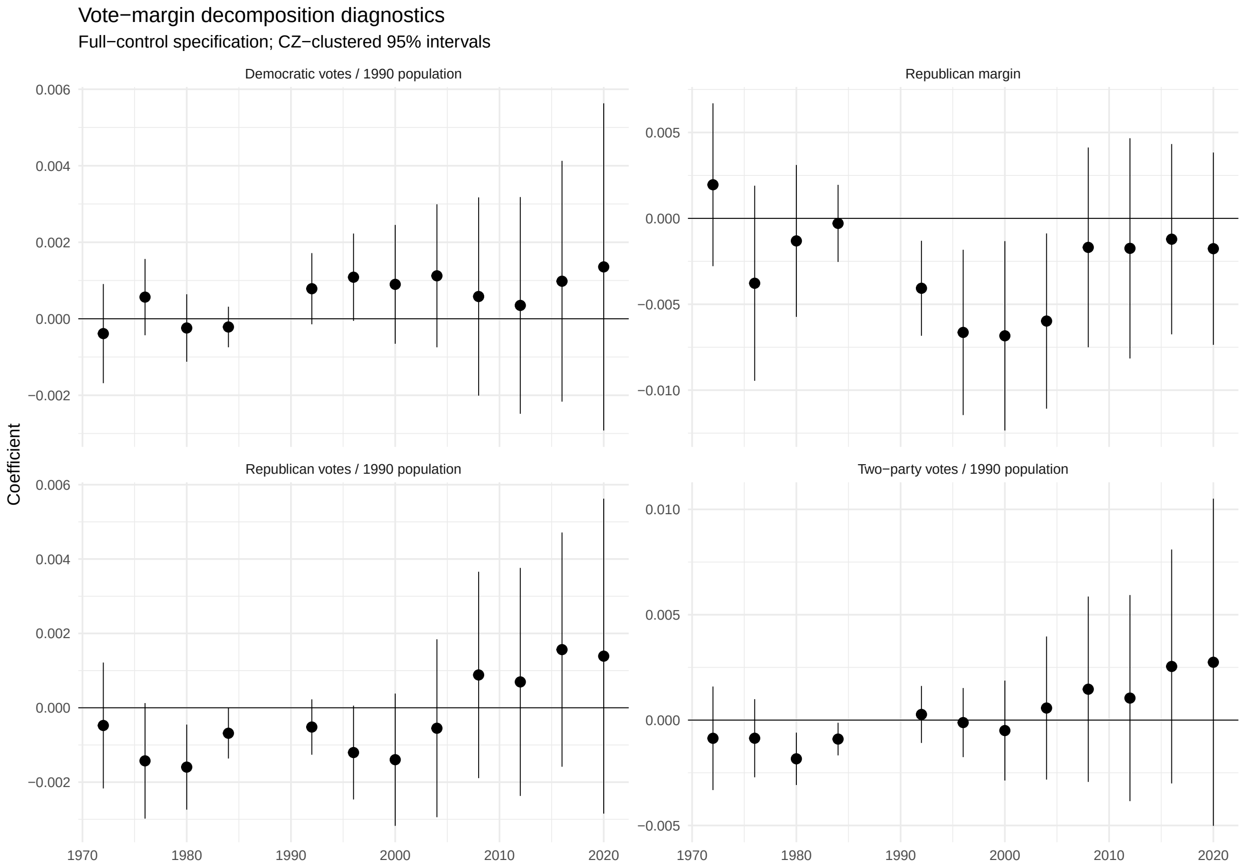


Figure 13: Outcome decomposition.

C Appendix C: Bartik Identification Diagnostics

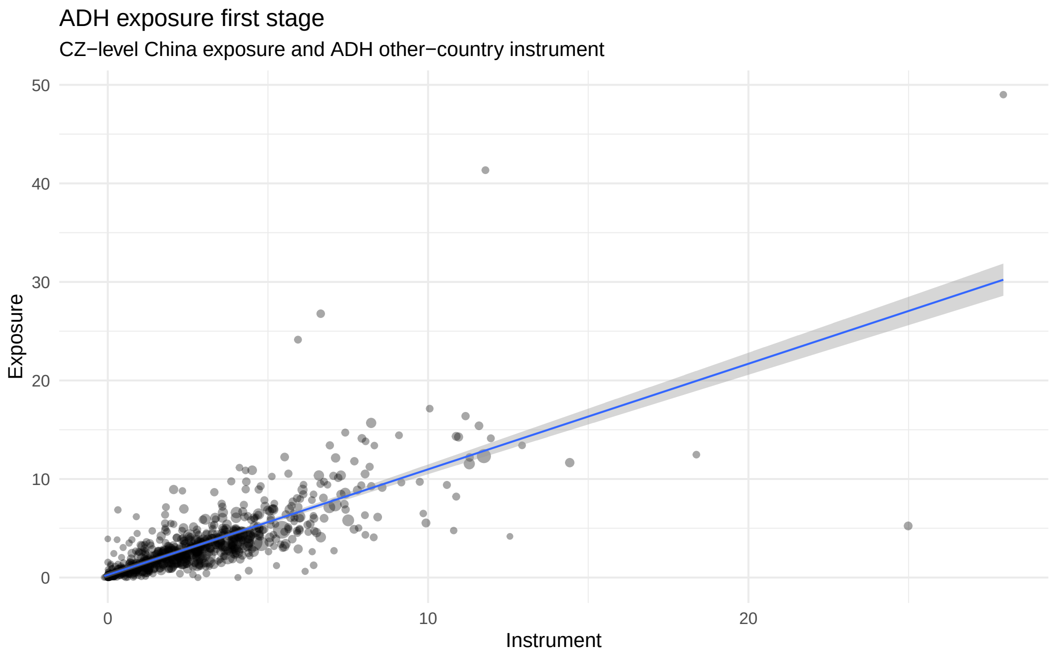


Figure 14: First-stage relationship between ADH exposure and the ADH instrument.

Table 5: Baseline balance: correlations with exposure and instrument.

Variable	Corr. with exposure	Corr. with instrument	Weighted mean	SD
l_shind_manuf_cbp	0.574	0.614	20.695	12.848
l_sh_popedu_c	-0.243	-0.287	48.021	8.574
l_sh_popfborn	-0.145	-0.131	10.146	4.969
l_sh_empl_f	0.028	0.006	63.624	6.634
l_sh_routine33	0.301	0.355	32.228	3.362
l_task_outsource	0.261	0.312	0.112	0.407
statefip	-0.085	-0.087	27.790	14.987

Table 6: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend_slope_rep_margin	minimal	exposure_1990_2007	0.0001	0.0002	0.601
pretrend_slope_rep_margin	core_controls	exposure_1990_2007	-0.0001	0.0002	0.700
pretrend_slope_rep_margin	full_controls	exposure_1990_2007	-0.0001	0.0002	0.548
pretrend_slope_rep_margin	minimal	instrument_1990_2007	0.0000	0.0002	0.976
pretrend_slope_rep_margin	core_controls	instrument_1990_2007	-0.0001	0.0002	0.556
pretrend_slope_rep_margin	full_controls	instrument_1990_2007	-0.0002	0.0002	0.498
pretrend_change_first_to_last	minimal	exposure_1990_2007	0.0075	0.0062	0.236

Table 6: Pre-1988 Republican-margin placebo diagnostics.

Outcome	Spec	Regressor	Estimate	Std. error	p-value
pretrend_change_first_to_last	core_controls	exposure_1990_2007	-0.0012	0.0056	0.833
pretrend_change_first_to_last	full_controls	exposure_1990_2007	-0.0018	0.0061	0.769
pretrend_change_first_to_last	minimal	instrument_1990_2007	0.0064	0.0083	0.445
pretrend_change_first_to_last	core_controls	instrument_1990_2007	-0.0001	0.0081	0.990
pretrend_change_first_to_last	full_controls	instrument_1990_2007	-0.0004	0.0087	0.959

Table 7: Industry-shift summary from ADH trade data diagnostics.

status	period	importer	sic87dd	start_imports	end_imports	import_growth
ok	1991-1999	OTH	3679	143699.73	2505878.0	2362178.3
ok	1991-1999	OTH	3577	40964.04	1565398.4	1524434.3
ok	1991-1999	OTH	3944	1285590.75	2732429.2	1446838.5
ok	1991-1999	OTH	3651	936563.69	2377214.8	1440651.1
ok	1991-1999	OTH	3861	58608.05	1004058.8	945450.8
ok	1991-1999	OTH	2369	1191708.38	2126562.8	934854.4
ok	1991-1999	OTH	2337	1214870.88	2095917.0	881046.1
ok	1991-1999	OTH	2091	235925.19	1114860.0	878934.8
ok	1991-1999	OTH	3161	623382.12	1466893.6	843511.5
ok	1991-1999	OTH	3089	256498.66	1082972.6	826474.0
ok	1991-1999	OTH	2325	634508.31	1300735.5	666227.2
ok	1991-1999	OTH	2392	605945.94	1227629.4	621683.4
ok	1991-1999	OTH	3021	374964.75	990012.8	615048.1
ok	1991-1999	OTH	2331	934854.62	1546733.8	611879.1
ok	1991-1999	OTH	3357	65468.17	666530.7	601062.5

D Appendix D: Subperiod Exposure and Dynamic Interpretation

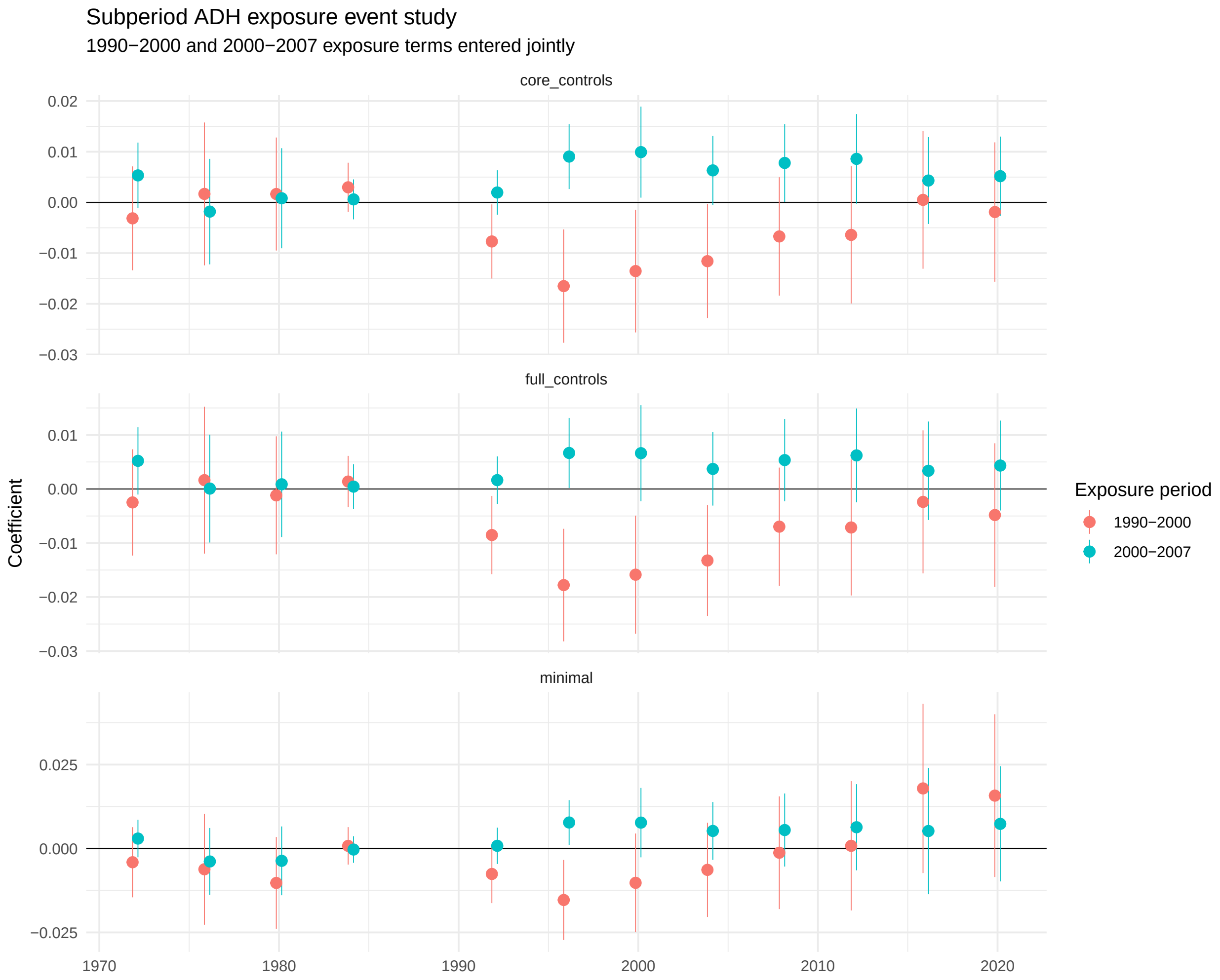


Figure 15: Subperiod exposure event-study estimates.

Table 8: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late	Approx. p-value
Minimal	1996	-0.0153	0.0077	-0.0231	0.001
Minimal	2000	-0.0102	0.0077	-0.0179	0.050
Minimal	2008	-0.0013	0.0055	-0.0068	0.508
Minimal	2016	0.0179	0.0052	0.0127	0.429
Minimal	2020	0.0157	0.0073	0.0084	0.579
Core controls	1996	-0.0165	0.0090	-0.0256	0.000
Core controls	2000	-0.0135	0.0099	-0.0235	0.002

Table 8: Selected horizons: early versus late ADH exposure.

Spec	Year	1990–2000 exposure	2000–2007 exposure	Early minus late	Approx. p-value
Core controls	2008	-0.0067	0.0078	-0.0145	0.042
Core controls	2016	0.0005	0.0043	-0.0038	0.642
Core controls	2020	-0.0019	0.0052	-0.0071	0.381
Full controls	1996	-0.0178	0.0067	-0.0245	0.000
Full controls	2000	-0.0159	0.0066	-0.0225	0.002
Full controls	2008	-0.0070	0.0053	-0.0123	0.070
Full controls	2016	-0.0024	0.0034	-0.0058	0.482
Full controls	2020	-0.0048	0.0043	-0.0092	0.252

E Appendix E: Standard-Error Diagnostics

Table 9: Variance-covariance diagnostics by specification and SE type.

sample	spec	se_type	status	vcov_repaired	repair_required	used_repaired	min_eigenvalue	negative_eigenvalues	max_abs_repair_change
main_1972_start	minimal_full_panel	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000022	0	NA
main_1972_start	minimal_full_panel	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
main_1972_start	minimal_full_panel	cluster_state	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
main_1972_start	minimal_full_panel	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0000403	10	NA
main_1972_start	minimal_full_panel	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000012	0	NA
main_1972_start	minimal_full_panel	driscoll_kraay_lag1	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
main_1972_start	minimal_full_panel	conley_250km	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
main_1972_start	minimal_full_panel	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0000057	2	NA
main_1972_start	minimal_full_panel	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0000041	4	NA
main_1972_start	minimal_full_panel	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0000073	4	NA
main_1972_start	interacted_core_controls_diagnostic	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000011	0	NA
main_1972_start	interacted_core_controls_diagnostic	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
main_1972_start	interacted_core_controls_diagnostic	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	1	NA
main_1972_start	interacted_core_controls_diagnostic	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0004383	35	NA
main_1972_start	interacted_core_controls_diagnostic	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000006	0	NA
main_1972_start	interacted_core_controls_diagnostic	driscoll_kraay_lag1	ok	FALSE	FALSE	FALSE	0.0000000	17	NA
main_1972_start	interacted_core_controls_diagnostic	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0000107	7	NA
main_1972_start	interacted_core_controls_diagnostic	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0000835	18	NA

Table 9: Variance-covariance diagnostics by specification and SE type.

sample	spec	se_type	status	vcov_repair_required	repair_required	used_repaired	min_eigenvalue	negative_eigenvalues	max_abs_repair_chan
main_1972_start	interacted_core_controls_diagnostic	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0001852	21	NA
main_1972_start	interacted_core_controls_diagnostic	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0003335	22	NA
main_1972_start	interacted_controls_full_panel	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000010	0	NA
main_1972_start	interacted_controls_full_panel	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
main_1972_start	interacted_controls_full_panel	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	18	NA
main_1972_start	interacted_controls_full_panel	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0006599	63	NA
main_1972_start	interacted_controls_full_panel	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000006	0	NA
main_1972_start	interacted_controls_full_panel	driscoll_kraay_lag4	ok	FALSE	FALSE	FALSE	0.0000000	35	NA
main_1972_start	interacted_controls_full_panel	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0000974	24	NA
main_1972_start	interacted_controls_full_panel	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0007051	36	NA
main_1972_start	interacted_controls_full_panel	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0011557	40	NA
main_1972_start	interacted_controls_full_panel	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0019790	40	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	20	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0019361	63	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	driscoll_kraay_lag4	ok	FALSE	FALSE	FALSE	0.0000000	35	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0000139	24	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0007107	36	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0003939	40	NA
main_1972_start	interacted_controls_full_panel_raw_controls_diagnostic	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0028599	40	NA
diagnostic_1952_start	minimal_full_panel	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000020	0	NA
diagnostic_1952_start	minimal_full_panel	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
diagnostic_1952_start	minimal_full_panel	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	minimal_full_panel	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0000385	14	NA
diagnostic_1952_start	minimal_full_panel	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000014	0	NA
diagnostic_1952_start	minimal_full_panel	driscoll_kraay_lag4	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	minimal_full_panel	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0000001	1	NA

Table 9: Variance-covariance diagnostics by specification and SE type.

sample	spec	se_type	status	vcov_repair_required	repair_required	vcov_repaired	min_eigenvalue	n_negative_eigenvalues	max_abs_repair_change
diagnostic_1952_start	minimal_full_panel	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0000112	4	NA
diagnostic_1952_start	minimal_full_panel	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0000233	7	NA
diagnostic_1952_start	minimal_full_panel	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0000152	7	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000014	0	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	10	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0004952	53	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000007	0	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	driscoll_kraay_lag1	ok	FALSE	FALSE	FALSE	0.0000000	25	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0000240	16	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0002296	28	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0004917	31	NA
diagnostic_1952_start	interacted_core_controls_diagnostic	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0010091	33	NA
diagnostic_1952_start	interacted_controls_full_panel	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000011	0	NA
diagnostic_1952_start	interacted_controls_full_panel	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	interacted_controls_full_panel	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	36	NA
diagnostic_1952_start	interacted_controls_full_panel	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0011482	94	NA
diagnostic_1952_start	interacted_controls_full_panel	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000006	0	NA
diagnostic_1952_start	interacted_controls_full_panel	driscoll_kraay_lag1	ok	FALSE	FALSE	FALSE	0.0000000	50	NA
diagnostic_1952_start	interacted_controls_full_panel	conley_250km	repair_required	TRUE	TRUE	FALSE	-0.0002214	39	NA
diagnostic_1952_start	interacted_controls_full_panel	conley_500km	repair_required	TRUE	TRUE	FALSE	-0.0012160	54	NA
diagnostic_1952_start	interacted_controls_full_panel	conley_750km	repair_required	TRUE	TRUE	FALSE	-0.0025146	56	NA
diagnostic_1952_start	interacted_controls_full_panel	conley_1000km	repair_required	TRUE	TRUE	FALSE	-0.0040057	57	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	heteroskedastic	ok	FALSE	FALSE	FALSE	0.0000001	0	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_cz	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_state	ok	FALSE	FALSE	FALSE	0.0000000	37	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	cluster_cz_year	repair_required	TRUE	TRUE	FALSE	-0.0040369	94	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	newey_west_lag1	ok	FALSE	FALSE	FALSE	0.0000000	0	NA
diagnostic_1952_start	interacted_controls_full_panel_raw_controls_diagnostic	driscoll_kraay_lag1	ok	FALSE	FALSE	FALSE	0.0000000	51	NA

Table 9: Variance-covariance diagnostics by specification and SE type.

sample	spec	se_type	status	vcov_repair	repair_required	used_repaired_[illegible]	min_eigenvalue	negative_eigenvalues	max_abs_repair_char
diagnostic	interacted_1952_start_controls_full_panel	conley_250km	repair_required_diagnostic	TRUE	TRUE	FALSE	-0.0000431	39	NA
diagnostic	interacted_1952_start_controls_full_panel	conley_500km	repair_required_diagnostic	TRUE	TRUE	FALSE	-0.0011073	54	NA
diagnostic	interacted_1952_start_controls_full_panel	conley_750km	repair_required_diagnostic	TRUE	TRUE	FALSE	-0.0009977	56	NA
diagnostic	interacted_1952_start_controls_full_panel	conley_1000km	repair_required_diagnostic	TRUE	TRUE	FALSE	-0.0055161	57	NA

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	note
diagnostic	interacted_1952_start	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	0.0000000	NA
diagnostic	interacted_1952_start	[illegible]	[illegible]	8	119	119	0	repair_required	TRUE	FALSE	-0.0011482	Two-way clustering includes very few year clusters; use as diagnostic only.
diagnostic	interacted_1952_start	[illegible]	[illegible]	0	119	47	72	ok	FALSE	FALSE	0.0000000	VCOV is rank deficient; use as diagnostic only unless justified.
diagnostic	interacted_1952_start	[illegible]	[illegible]	1	119	119	0	repair_required	TRUE	FALSE	-0.0040057	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	interacted_1952_start	[illegible]	[illegible]	0	119	119	0	repair_required	TRUE	FALSE	-0.0002214	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	diagnostic_note
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	repair_required	TRUE	FALSE	-0.0012160	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	1	119	119	0	repair_required	TRUE	FALSE	-0.0025146	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	0	119	51	68	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	8	119	119	0	repair_required	TRUE	FALSE	-0.0040369	Two-way clustering includes very few year clusters; use as diagnostic only.
diagnostic	[illegible]	[illegible]	[illegible]	0	119	50	69	ok	FALSE	FALSE	[illegible]	VCOV is rank deficient; use as diagnostic only unless justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	event	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	note
diagnostic	[illegible]	[illegible]	[illegible]	1	119	119	0	diagnostic	repair_required	TRUE	FALSE	0.005516	[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	diagnostic	repair_required	TRUE	FALSE	0.000043	[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	0	119	119	0	diagnostic	repair_required	TRUE	FALSE	0.001107	[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	1	119	119	0	diagnostic	repair_required	TRUE	FALSE	0.000997	[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	diagnostic_note
[illegible]	[illegible]	[illegible]	[illegible]	0	119	63	56	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
[illegible]	[illegible]	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	[illegible]	NA
[illegible]	[illegible]	[illegible]	[illegible]	0	119	119	0	ok	FALSE	FALSE	[illegible]	NA
[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	ok	FALSE	FALSE	[illegible]	NA
[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	repair_required	TRUE	FALSE	0.0004952	Two-way clustering includes very few year clusters; use as diagnostic only.
[illegible]	[illegible]	[illegible]	[illegible]	0	68	47	21	ok	FALSE	FALSE	[illegible]	VCOV is rank deficient; use as diagnostic only unless justified.
[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	repair_required	TRUE	FALSE	0.0010091	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	repair_required	TRUE	FALSE	0.0000240	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	event	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	diagnostic_note
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	repair_required	TRUE	FALSE	0.0002296	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	1	68	68	0	repair_required	TRUE	FALSE	0.0004917	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	68	28	40	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	68	68	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	17	17	0	ok	FALSE	FALSE	[illegible]	NA
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	7	17	17	0	repair_required	TRUE	FALSE	0.0000385	Two-way clustering includes very few year clusters; use as diagnostic only.
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	17	17	0	ok	FALSE	FALSE	[illegible]	NA

Table 10: SE warning diagnostics.

sample	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	note
diagnostic	[illegible]	0	17	17	0	repair_required	TRUE	FALSE	-0.0000152	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	0	17	17	0	repair_required	TRUE	FALSE	-0.0000001	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	0	17	17	0	repair_required	TRUE	FALSE	-0.0000112	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	0	17	17	0	repair_required	TRUE	FALSE	-0.0000233	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
diagnostic	[illegible]	0	17	17	0	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs; do not report without further diagnosis.
diagnostic	[illegible]	0	17	17	0	ok	FALSE	FALSE	[illegible]	NA

Table 10: SE warning diagnostics.

sample	spec	se_type	event	[illegible]									note
diagnostic	[illegible]	[illegible]	[illegible]	[illegible]	0	17	17	0	ok	FALSE	FALSE	[illegible]	NA
main_	[illegible]	[illegible]	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	[illegible]	NA
main_	[illegible]	[illegible]	[illegible]	[illegible]	5	84	84	0	repair_required	TRUE	FALSE	0.0006599	Two-way clustering includes very few year clusters; use as diagnostic only.
main_	[illegible]	[illegible]	[illegible]	[illegible]	0	84	47	37	ok	FALSE	FALSE	[illegible]	VCOV is rank deficient; use as diagnostic only unless justified.
main_	[illegible]	[illegible]	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	0.0019790	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_	[illegible]	[illegible]	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	0.0000974	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_	[illegible]	[illegible]	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	0.0007051	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	note
main_interact	[illegible]	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	0.0011557	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact	[illegible]	[illegible]	[illegible]	0	84	26	58	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
main_interact	[illegible]	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	[illegible]	NA
main_interact	[illegible]	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	[illegible]	NA
main_interact	[illegible]	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	[illegible]	NA
main_interact	[illegible]	[illegible]	[illegible]	5	84	84	0	repair_required	TRUE	FALSE	0.0019361	Two-way clustering includes very few year clusters; use as diagnostic only.
main_interact	[illegible]	[illegible]	[illegible]	0	84	52	32	ok	FALSE	FALSE	[illegible]	VCOV is rank deficient; use as diagnostic only unless justified.
main_interact	[illegible]	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	0.0028599	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	note
main_interact	1972_start_controls	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	-0.0000139		Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact	1972_start_controls	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	-0.0007107		Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact	1972_start_controls	[illegible]	[illegible]	0	84	84	0	repair_required	TRUE	FALSE	-0.0003939		Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact	1972_start_controls	driscoll_kraay	[illegible]	0	84	15	69	ok	FALSE	FALSE	0.0000		Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
main_interact	1972_start_controls	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	0.0001		NA
main_interact	1972_start_controls	[illegible]	[illegible]	0	84	84	0	ok	FALSE	FALSE	0.0000		NA
main_interact	1972_start_controls	cluster_12	[illegible]	0	48	48	0	ok	FALSE	FALSE	0.0000		NA

Table 10: SE warning diagnostics.

sample spec se_type event_min_coef_... (overlapped headers)								diagnostic_note
main_interact_start 1972_cluster12cz_year controls_diagnostic 0.0010654 0.0027766	5	48	48	0	repair_required	TRUE	FALSE 0.0004383	Two-way clustering includes very few year clusters; use as diagnostic only.
main_interact_start 1972_cluster12state controls_diagnostic 0.0012830 0.0037894	0	48	48	0	ok	FALSE	FALSE 0.0000	NA
main_interact_start 1972_conley12100km controls_diagnostic 0.0034265 0.0056664	0	48	48	0	repair_required	TRUE	FALSE 0.0003335	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact_start 1972_conley12250km controls_diagnostic 0.0037304 0.0040624	0	48	48	0	repair_required	TRUE	FALSE 0.0000107	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact_start 1972_conley12500km controls_diagnostic 0.0046573 0.0043840	0	48	48	0	repair_required	TRUE	FALSE 0.0000835	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_interact_start 1972_conley12750km controls_diagnostic 0.0012407 0.0047087	0	48	48	0	repair_required	TRUE	FALSE 0.0001852	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	event	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]000	0	48	13	35	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs and rank-deficient VCOV; do not report without further diagnosis.
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]685	0	48	48	0	ok	FALSE	FALSE	0.0000[illegible]11	NA
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]018	0	48	48	0	ok	FALSE	FALSE	0.0000[illegible]06	NA
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]716	0	12	12	0	ok	FALSE	FALSE	0.0000[illegible]01	NA
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]529	6	12	12	0	repair[illegible]	TRUE	FALSE	0.0000[illegible]	Two-way clustering includes very few year clusters; use as diagnostic only.
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]696	0	12	12	0	ok	FALSE	FALSE	0.0000[illegible]01	NA
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]565	0	12	12	0	repair[illegible]	TRUE	FALSE	0.0000[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]180	0	12	12	0	ok	FALSE	FALSE	0.0000[illegible]01	NA
main_[illegible]	1972[illegible]	[illegible]	[illegible]	[illegible]483	0	12	12	0	repair[illegible]	TRUE	FALSE	0.0000[illegible]	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.

Table 10: SE warning diagnostics.

sample	spec	se_type	[illegible]								diagnostic_note
main_[illegible]	1972_[illegible]	conley_[illegible]	[illegible]460	0	12	12	0	repair_required	TRUE	FALSE	-0.0000041	Conley VCOV is non-positive-definite before repair or requires numerical repair; diagnostic only unless separately justified.
main_[illegible]	1972_[illegible]	driscoll_kraay_[illegible]	[illegible]000	0	12	12	0	ok	FALSE	FALSE	[illegible]	Suspicious near-zero Driscoll-Kraay SEs; do not report without further diagnosis.
main_[illegible]	1972_[illegible]	[illegible]	[illegible]428	0	12	12	0	ok	FALSE	FALSE	[illegible]22	NA
main_[illegible]	1972_[illegible]	newey_west_[illegible]	[illegible]769	0	12	12	0	ok	FALSE	FALSE	[illegible]12	NA

F Appendix F: Exploratory Mechanism Evidence

F.1 NaNDA turnout, vote ratios, and partisanship

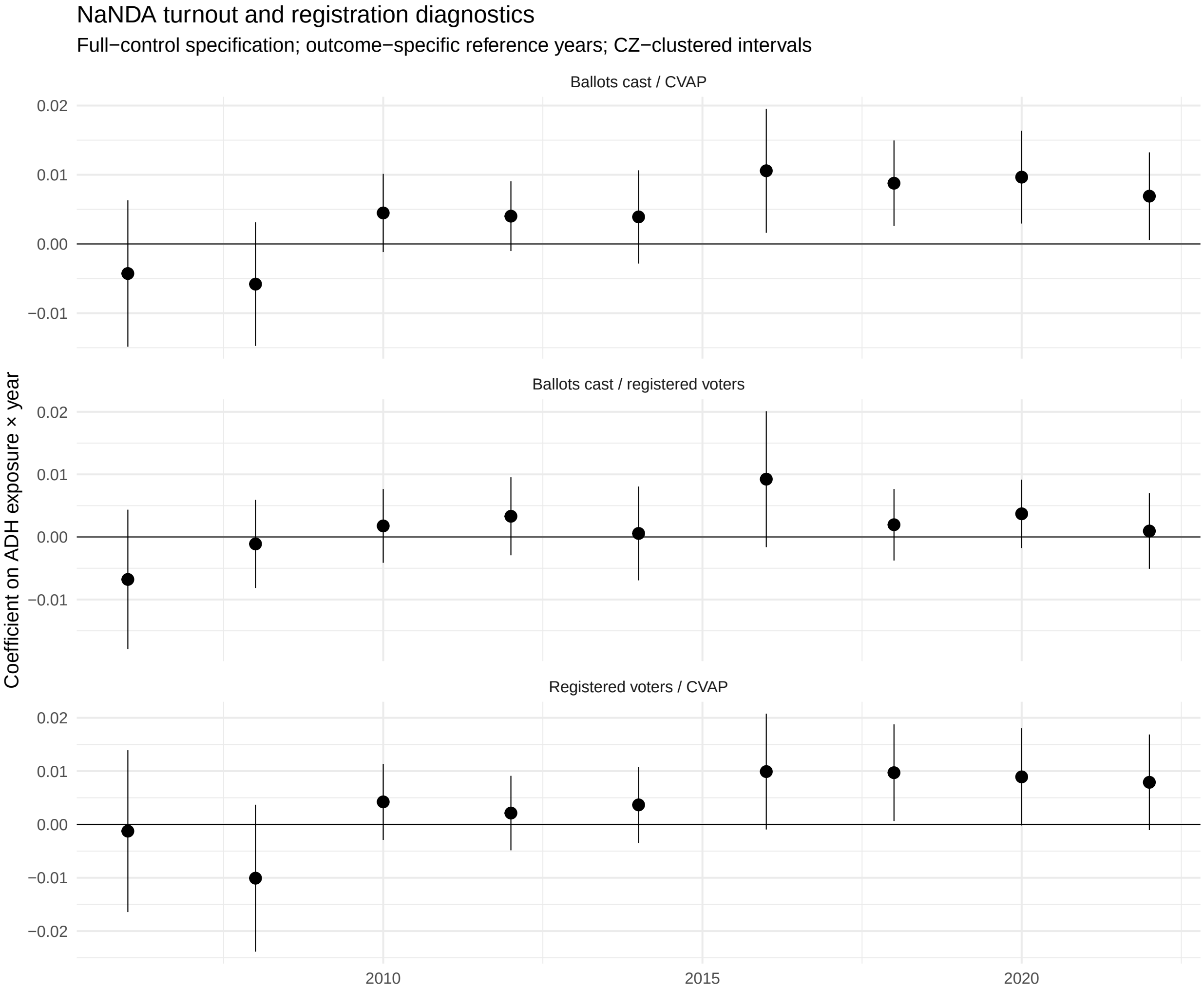


Figure 16: NaNDA turnout and registration outcomes.

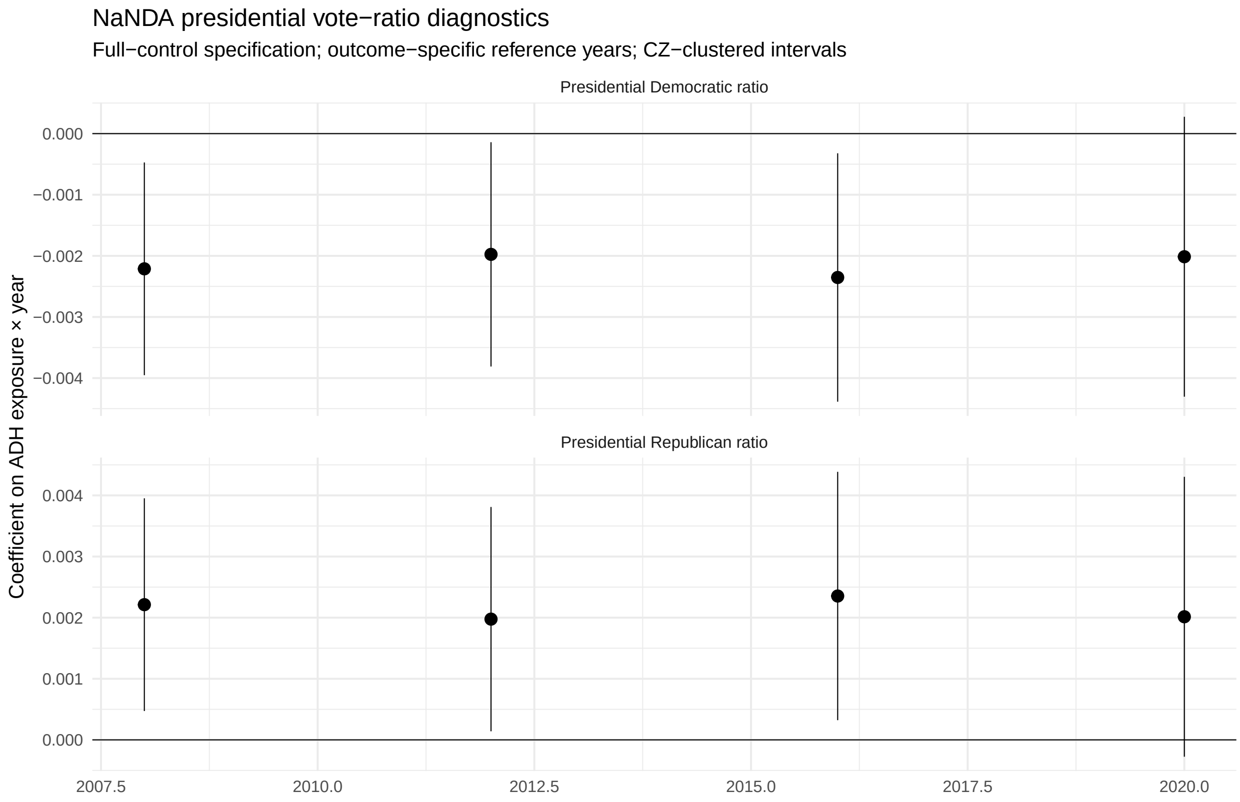


Figure 17: NaNDA presidential vote ratios.

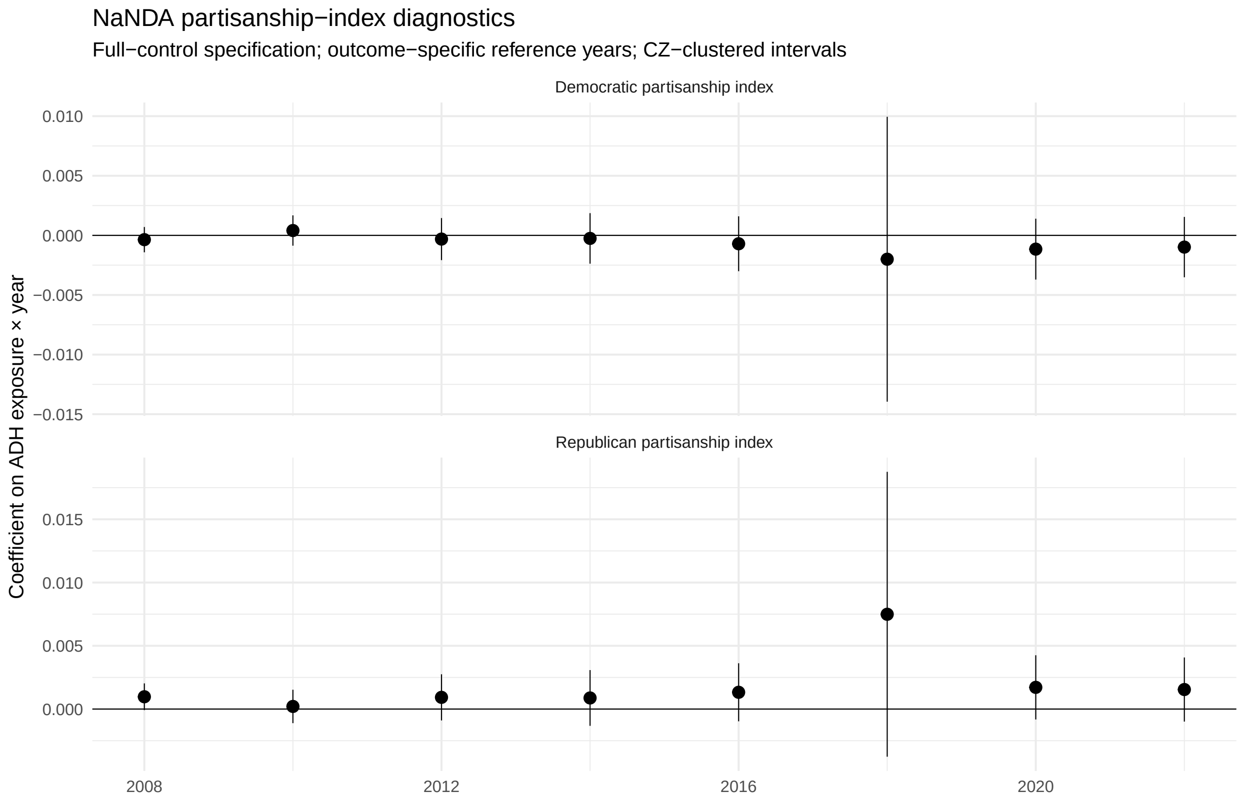


Figure 18: NaNDA partisanship indices.

Table 11: NaNDA outcome availability by year.

Outcome	Year	Nonmissing CZs	Zero CZs	Available
partisan_index_dem	2004	722	722	FALSE
partisan_index_dem	2006	722	0	TRUE
partisan_index_dem	2008	722	0	TRUE
partisan_index_dem	2010	722	0	TRUE
partisan_index_dem	2012	722	0	TRUE
partisan_index_dem	2014	722	0	TRUE
partisan_index_dem	2016	722	0	TRUE
partisan_index_dem	2018	722	253	TRUE
partisan_index_dem	2020	722	0	TRUE
partisan_index_dem	2022	722	0	TRUE
partisan_index_rep	2004	722	722	FALSE
partisan_index_rep	2006	722	0	TRUE
partisan_index_rep	2008	722	0	TRUE
partisan_index_rep	2010	722	0	TRUE
partisan_index_rep	2012	722	0	TRUE
partisan_index_rep	2014	722	0	TRUE
partisan_index_rep	2016	722	0	TRUE
partisan_index_rep	2018	722	253	TRUE

Table 11: NaNDA outcome availability by year.

Outcome	Year	Nonmissing CZs	Zero CZs	Available
partisan_index_rep	2020	722	0	TRUE
partisan_index_rep	2022	722	0	TRUE
pres_dem_ratio	2004	722	0	TRUE
pres_dem_ratio	2006	0	0	FALSE
pres_dem_ratio	2008	722	0	TRUE
pres_dem_ratio	2010	0	0	FALSE
pres_dem_ratio	2012	722	0	TRUE
pres_dem_ratio	2014	0	0	FALSE
pres_dem_ratio	2016	722	0	TRUE
pres_dem_ratio	2018	0	0	FALSE
pres_dem_ratio	2020	722	0	TRUE
pres_dem_ratio	2022	0	0	FALSE
pres_rep_ratio	2004	722	0	TRUE
pres_rep_ratio	2006	0	0	FALSE
pres_rep_ratio	2008	722	0	TRUE
pres_rep_ratio	2010	0	0	FALSE
pres_rep_ratio	2012	722	0	TRUE
pres_rep_ratio	2014	0	0	FALSE
pres_rep_ratio	2016	722	0	TRUE
pres_rep_ratio	2018	0	0	FALSE
pres_rep_ratio	2020	722	0	TRUE
pres_rep_ratio	2022	0	0	FALSE
reg_voter_turnout_pct	2004	717	0	TRUE
reg_voter_turnout_pct	2006	661	40	TRUE
reg_voter_turnout_pct	2008	670	2	TRUE
reg_voter_turnout_pct	2010	703	0	TRUE
reg_voter_turnout_pct	2012	700	1	TRUE
reg_voter_turnout_pct	2014	702	16	TRUE
reg_voter_turnout_pct	2016	699	61	TRUE
reg_voter_turnout_pct	2018	701	1	TRUE
reg_voter_turnout_pct	2020	702	0	TRUE
reg_voter_turnout_pct	2022	704	0	TRUE
reg_voters_pct	2004	721	0	TRUE
reg_voters_pct	2006	722	50	TRUE
reg_voters_pct	2008	721	43	TRUE
reg_voters_pct	2010	721	15	TRUE
reg_voters_pct	2012	720	16	TRUE
reg_voters_pct	2014	721	16	TRUE
reg_voters_pct	2016	720	17	TRUE
reg_voters_pct	2018	721	16	TRUE
reg_voters_pct	2020	720	15	TRUE
reg_voters_pct	2022	721	15	TRUE
voter_turnout_pct	2004	722	0	TRUE
voter_turnout_pct	2006	722	62	TRUE

Table 11: NaNDA outcome availability by year.

Outcome	Year	Nonmissing CZs	Zero CZs	Available
voter_turnout_pct	2008	722	14	TRUE
voter_turnout_pct	2010	722	0	TRUE
voter_turnout_pct	2012	722	2	TRUE
voter_turnout_pct	2014	722	17	TRUE
voter_turnout_pct	2016	722	63	TRUE
voter_turnout_pct	2018	721	1	TRUE
voter_turnout_pct	2020	721	0	TRUE
voter_turnout_pct	2022	721	0	TRUE

F.2 ADHM 2020 public political-supply diagnostics

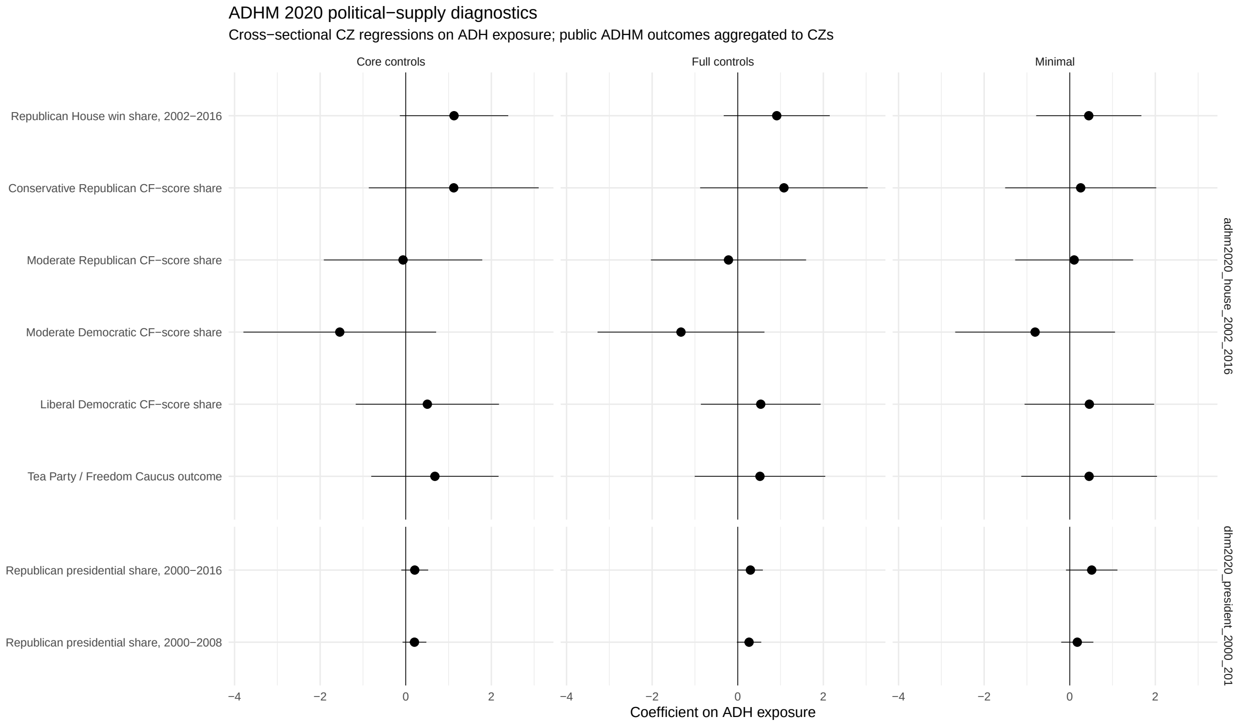


Figure 19: ADHM 2020 public political-supply mechanism regressions.

Table 12: ADHM 2020 mechanism regressions.

Outcome	Spec	Estimate	Std. error	p-value	N
Republican House win share, 2002-2016	Minimal	0.445	0.627	0.482	722
Republican House win share, 2002-2016	Core controls	1.128	0.646	0.087	722
Republican House win share, 2002-2016	Full controls	0.911	0.632	0.156	722
Conservative Republican CF-score share	Minimal	0.256	0.901	0.778	722
Conservative Republican CF-score share	Core controls	1.122	1.013	0.274	722

Table 12: ADHM 2020 mechanism regressions.

Outcome	Spec	Estimate	Std. error	p-value	N
Conservative Republican CF-score share	Full controls	1.080	0.999	0.285	722
Moderate Republican CF-score share	Minimal	0.103	0.703	0.884	722
Moderate Republican CF-score share	Core controls	-0.063	0.944	0.947	722
Moderate Republican CF-score share	Full controls	-0.216	0.925	0.816	722
Moderate Democratic CF-score share	Minimal	-0.809	0.953	0.400	722
Moderate Democratic CF-score share	Core controls	-1.541	1.149	0.186	722
Moderate Democratic CF-score share	Full controls	-1.326	0.995	0.189	722
Liberal Democratic CF-score share	Minimal	0.458	0.772	0.556	722
Liberal Democratic CF-score share	Core controls	0.505	0.854	0.557	722
Liberal Democratic CF-score share	Full controls	0.538	0.714	0.455	722
Tea Party / Freedom Caucus outcome	Minimal	0.453	0.809	0.578	722
Tea Party / Freedom Caucus outcome	Core controls	0.681	0.759	0.374	722
Tea Party / Freedom Caucus outcome	Full controls	0.520	0.778	0.508	722
Republican presidential share, 2000-2016	Minimal	0.514	0.307	0.101	722
Republican presidential share, 2000-2016	Core controls	0.210	0.160	0.196	722
Republican presidential share, 2000-2016	Full controls	0.296	0.148	0.052	722
Republican presidential share, 2000-2008	Minimal	0.176	0.192	0.362	722
Republican presidential share, 2000-2008	Core controls	0.204	0.142	0.159	722
Republican presidential share, 2000-2008	Full controls	0.265	0.145	0.075	722

G Appendix G: Replication Notes

The complete analysis can be regenerated from the project root with:

```r
source(here::here("src", "run_pipeline.R"))
```

For the final appendix-style crosswalk-sensitivity run:

```r
Sys.setenv(RUN_CROSSWALK_SENSITIVITY = "true")
source(here::here("src", "run_pipeline.R"))
```

The replication-code PDF can be generated from the `report` folder with:

```r
source("make_replication_code_pdf.R")
```